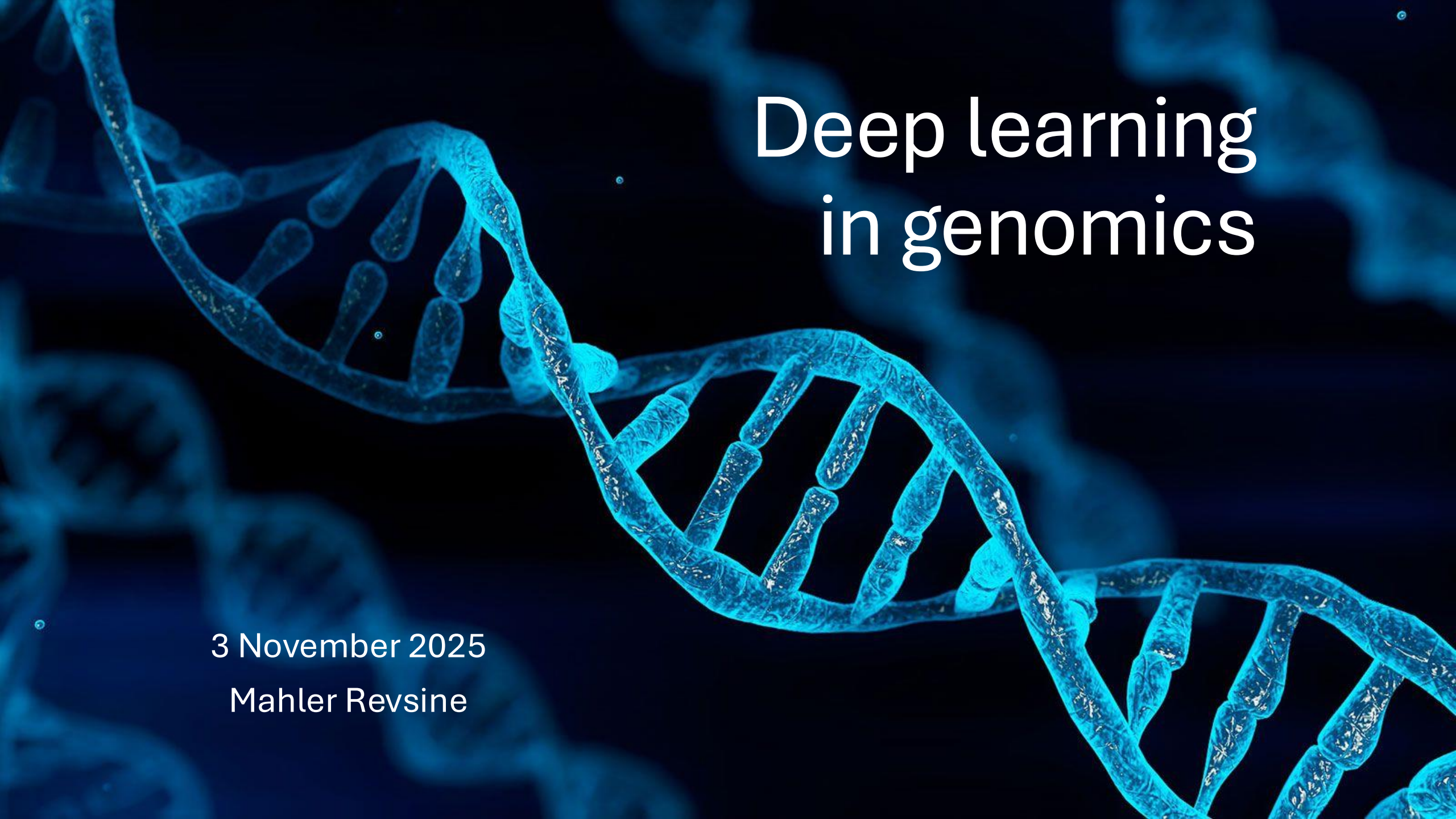




Deep learning in genomics

3 November 2025
Mahler Revsine

Announcements

- Assignment 4 grades released
 - Great scores overall! 😊
- Preliminary project report due tonight
- Assignment 5 due Friday night
 - Last chance to use late days

Deep Learning in Genomics Journal Club

- Discussion group for students interested in genomic deep learning
- We meet every other Tuesday at 12pm
- Participation is optional, feel free to just sit and listen
- Next meetings 11/4 and 11/18
- Email me if interested! All are welcome



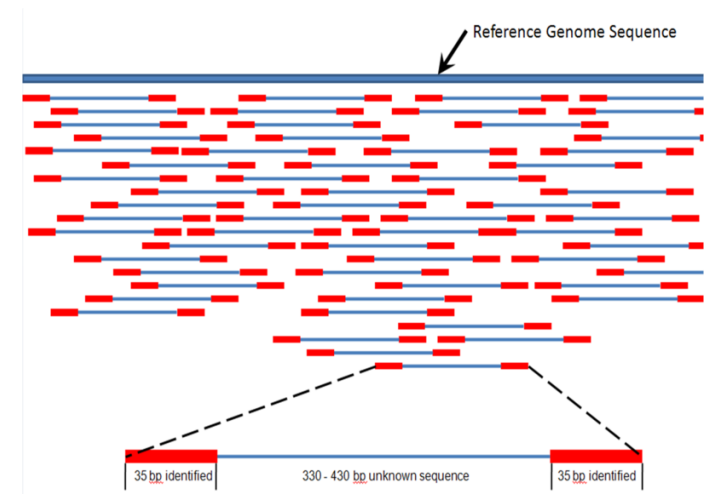
Imagine you are a doctor treating a patient with some condition



You sequence their DNA...



...align it to the reference genome...



...and find all of their variants

```
##fileformat=VCFv4.1
##fileDate=20140930
##source=23andme2vcf.pl https://github.com/arrogantrobot/23
##reference=file://23andme_v3_hg19_ref.txt.gz
##FORMAT=<ID=GT,Number=1,Type=String,Description="Genotype"
#CHROM POS ID REF ALT QUAL FILTER INFO FORMAT GEN
chr1 82154 rs4477212 a . . . . GT 0/0
chr1 752566 rs3094315 g A . . . . GT 1/1
chr1 752721 rs3131972 A G . . . . GT 1/1
chr1 798959 rs11240777 g . . . . . GT 0/0
chr1 800007 rs6681049 T C . . . . GT 1/1
chr1 838555 rs4970383 c . . . . . GT 0/0
chr1 846808 rs4475691 C . . . . . GT 0/0
chr1 854250 rs7537756 A . . . . . GT 0/0
chr1 861808 rs13302982 A G . . . . GT 1/1
chr1 873558 rs1110052 G T . . . . GT 1/1
chr1 882033 rs2272756 G A . . . . GT 0/1
chr1 888659 rs3748597 T C . . . . GT 1/1
chr1 891945 rs13303106 A G . . . . GT 0/1
```

How do you determine which variant(s) are responsible for their condition?

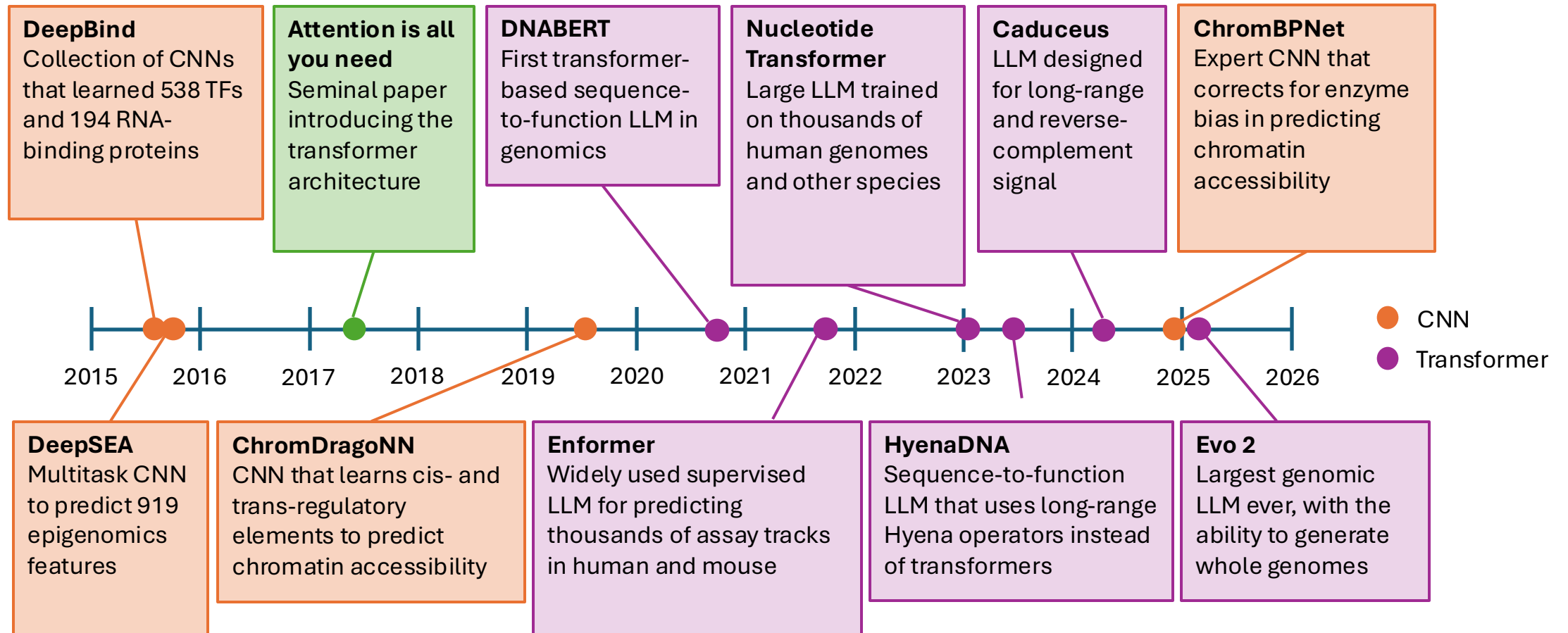
What does a DNA variant do?

Method	Drawback(s)
Query it in a database	The variant might not exist in a database The variant might act differently under different conditions
Assume loss-of-function of nearby gene	Not always an accurate assumption Function of nearby gene may not itself be understood
Genome-Wide Association Study (GWAS)	Not applicable for variants that only exist in a single person Underpowered for rare variants that only occur in a few people Relies on undersized datasets containing a small number of SNPs Requires rigorous statistical testing Can only model one variant at a time Assumes additive linear effects of variants Difficult to adapt to a polygenic/omnigenic model of traits In practice, only explains a fraction of heritability

Variant effect prediction

- What **effect** does **variant** X have on us? What **trait** does it create?
- Arguably the most important task in bioinformatics
- Historically studied using Genome-Wide Association Studies (**GWAS**) in connection with Quantitative Trait Loci (**QTL**) analysis
 - This approach has many drawbacks
- Could be studied using deep learning

Deep learning is now widespread in genomics



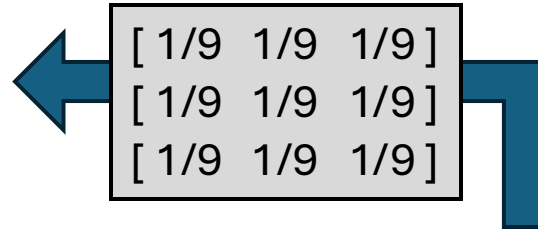
Deep learning basics

- Learn **patterns** from **some data** that can generalize to **all data**
- 3 key architectural pieces in genomics
 - Convolutions
 - Feed forward networks
 - Attention

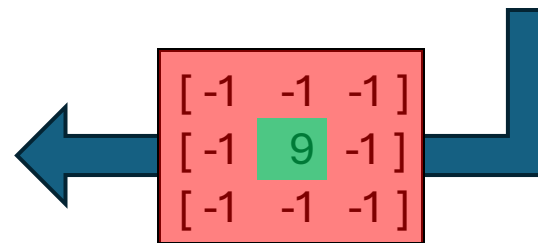
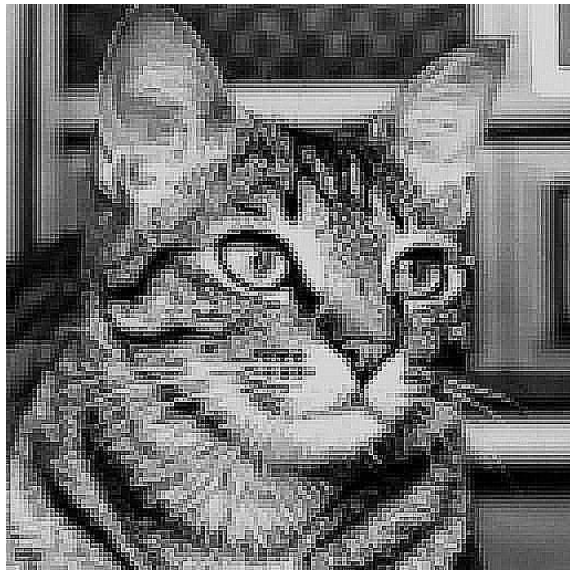
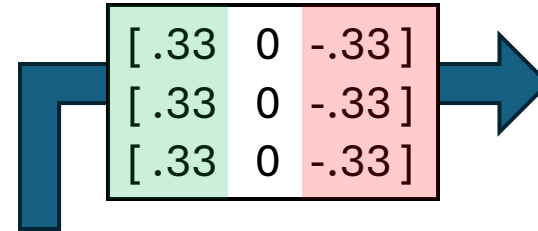


CONVOLUTIONS

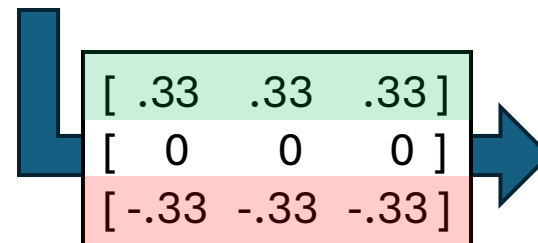
Blur



Vertical lines



Sharpen



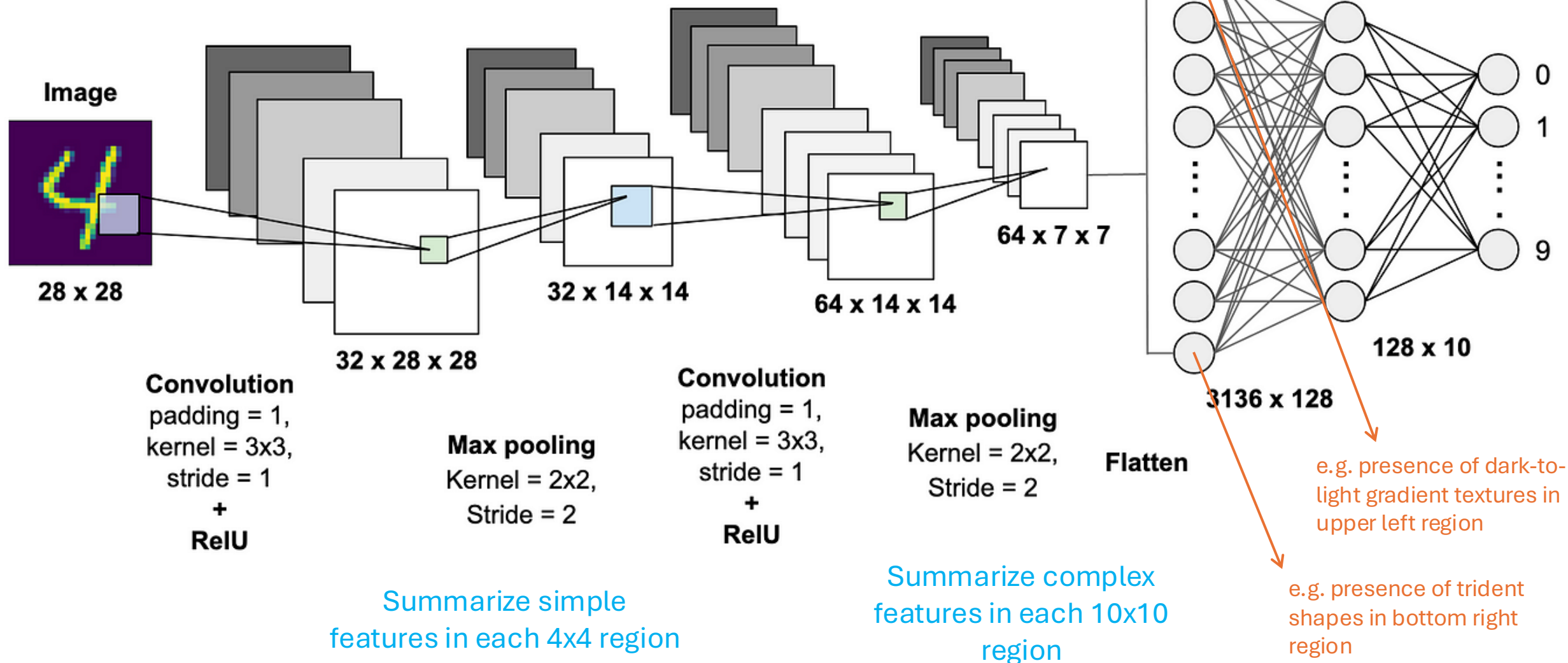
Horizontal lines

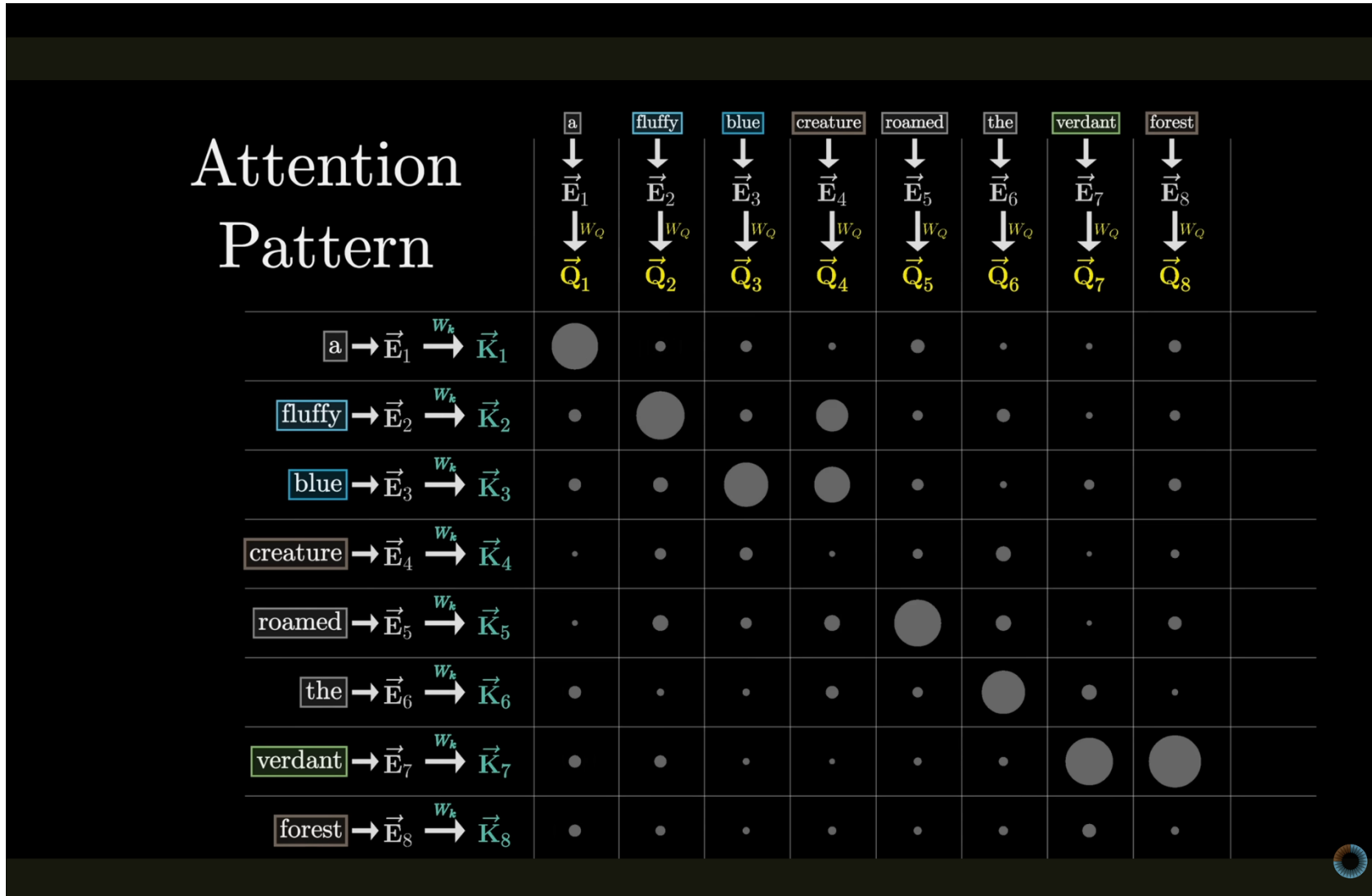


In each 3x3 region: Simple features e.g. vertical lines, horizontal lines, etc. (32 total)

In each 8x8 region: Complex features e.g. patterns, textures, combinations of shapes, etc. (64 total)

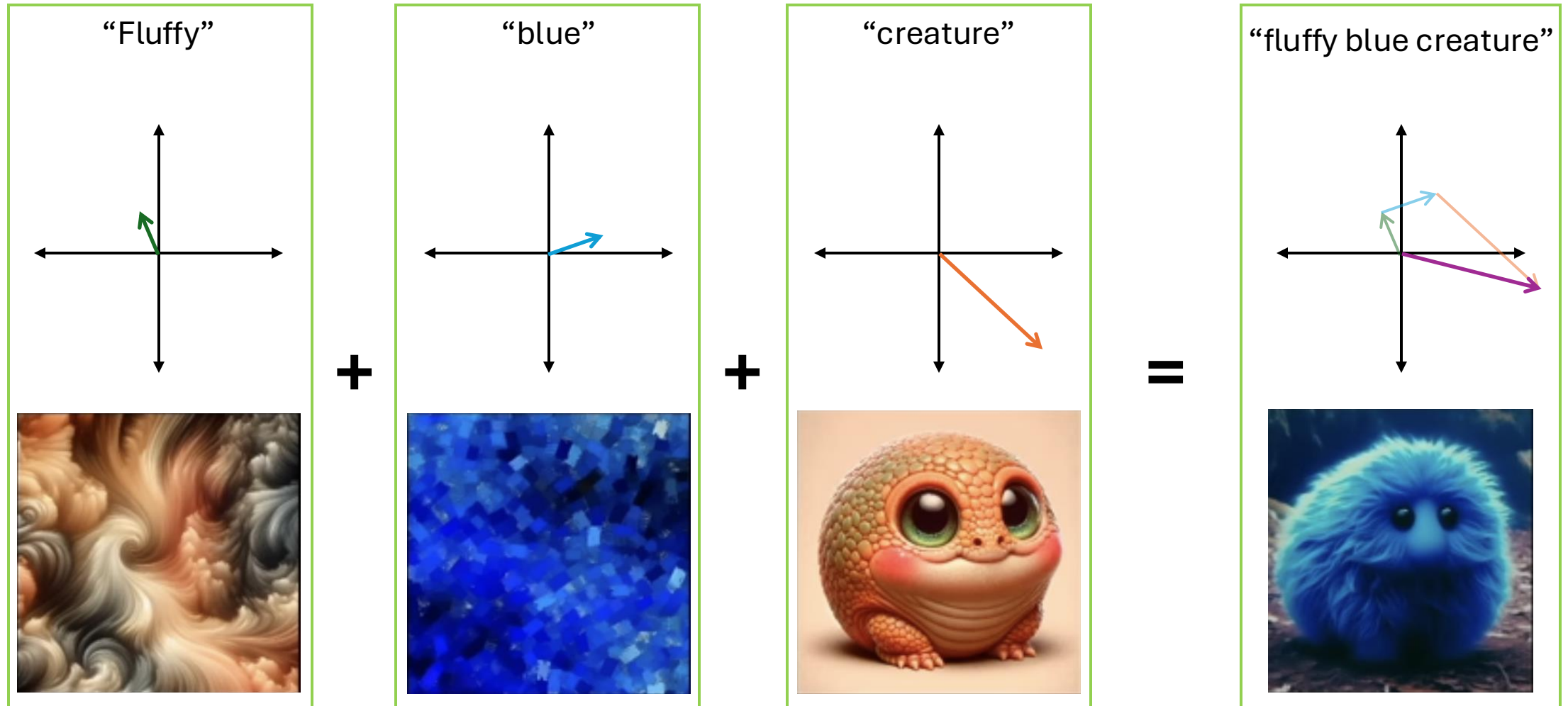
Learn complex patterns between features and labels





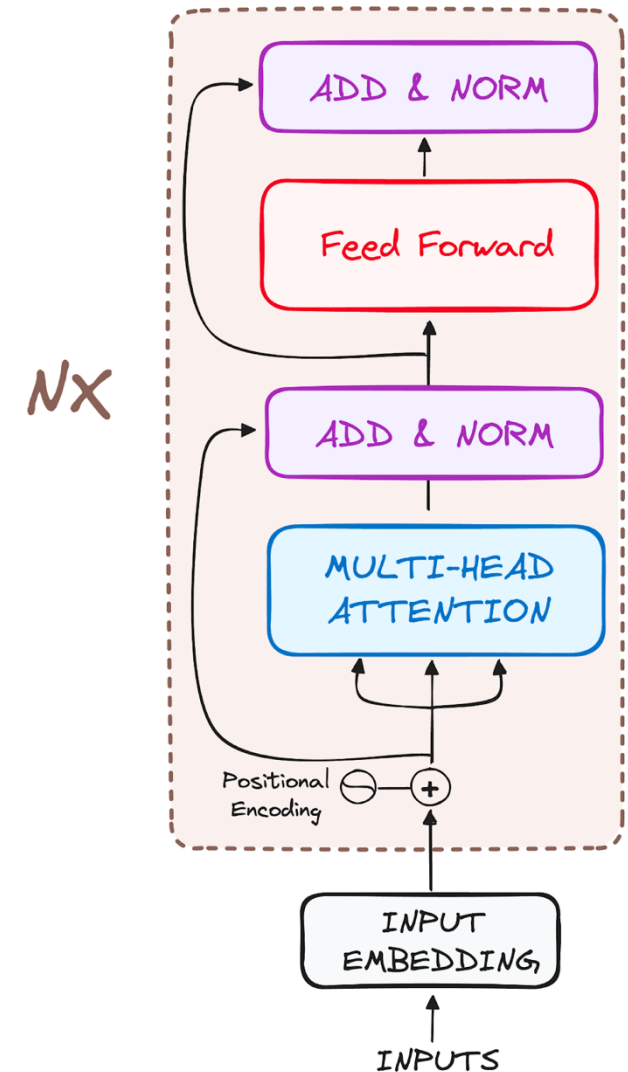
Attention tells us which words **relate** to which other words

Attention **modifies** the **vector representation** of a **word** to include its **additional meaning**

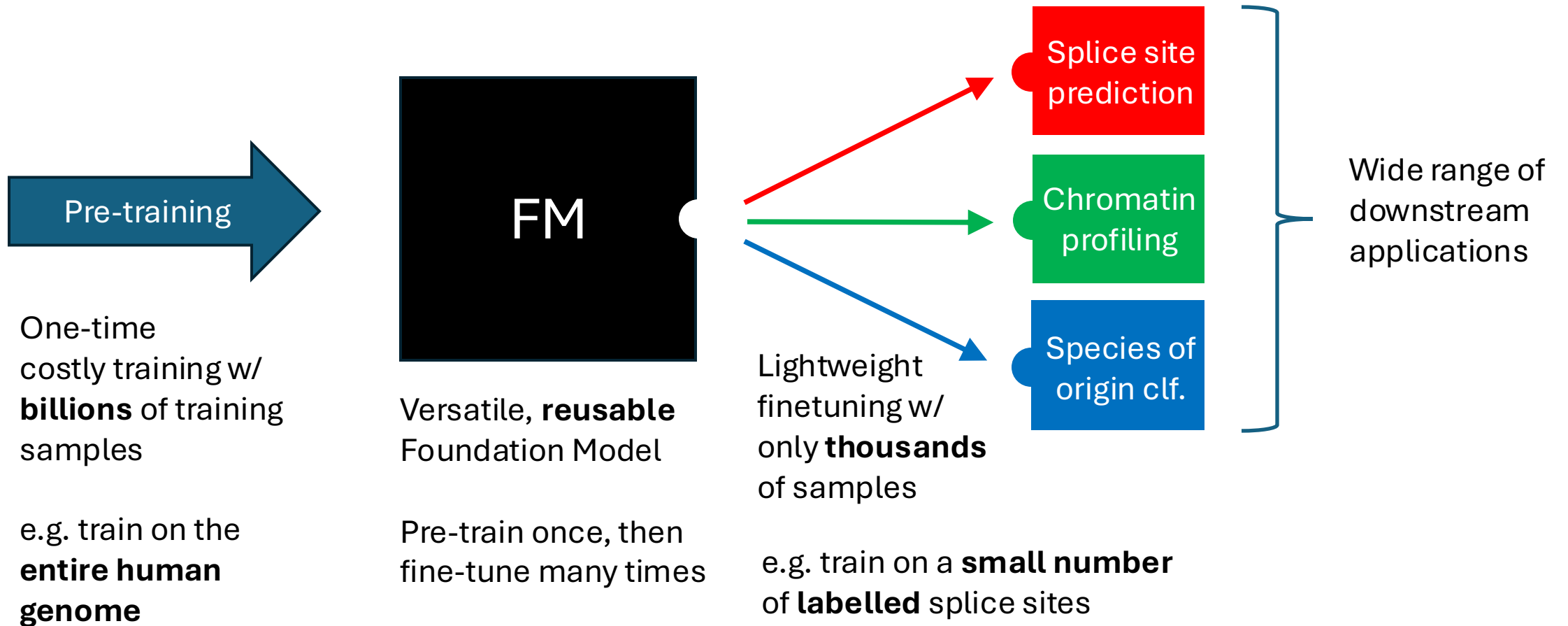


Why use LLMs in genomics?

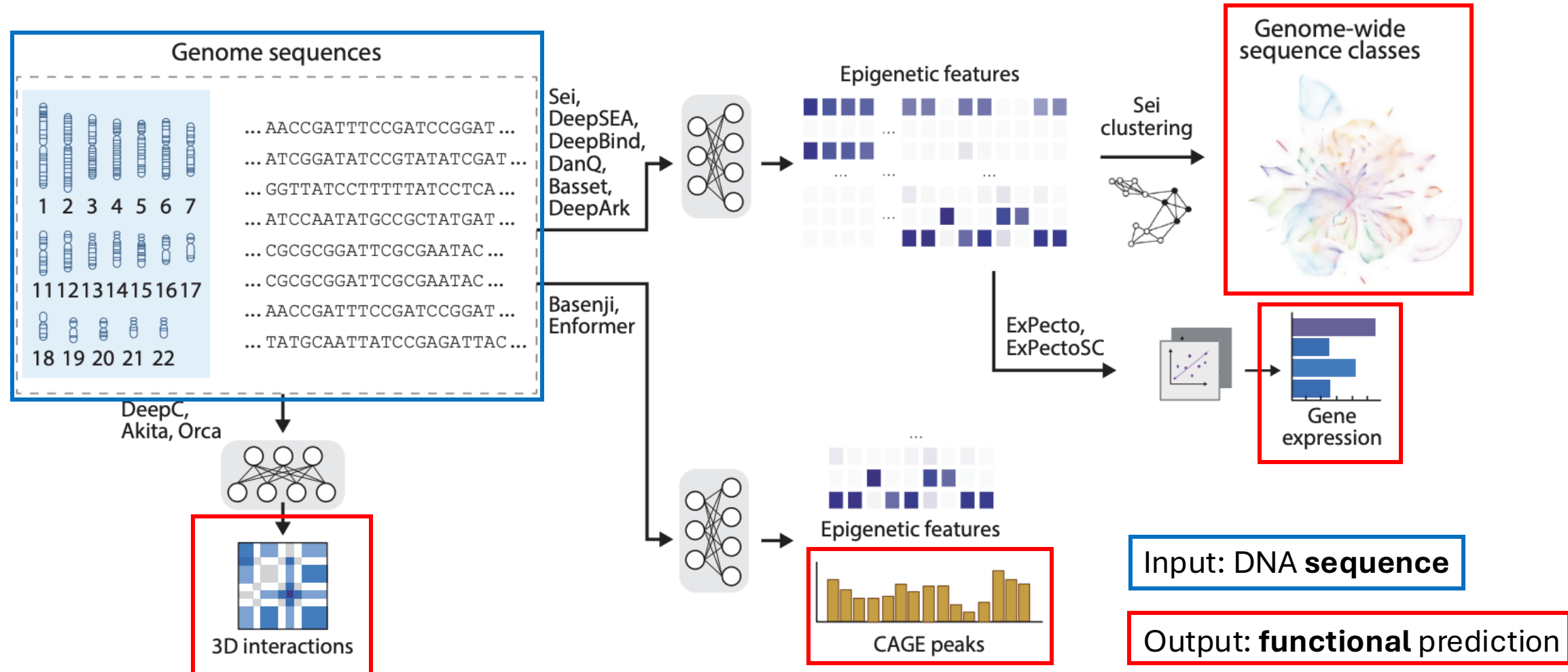
- LLMs employ the most advanced **architecture** ever made for AI
 - Core advancement is **Attention**
 - First truly effective way to process **sequential** data like **text**
 - Attention allows models to get really **big**, and bigger is better
- Operate at **lower cost** than humans
- Can conduct massively **parallel** experiments with **little human effort**
- Can model **rare** or **novel sequences**



Foundation model (FM)



Sequence-to-function models



Enformer

Article | [Open access](#) | Published: 04 October 2021

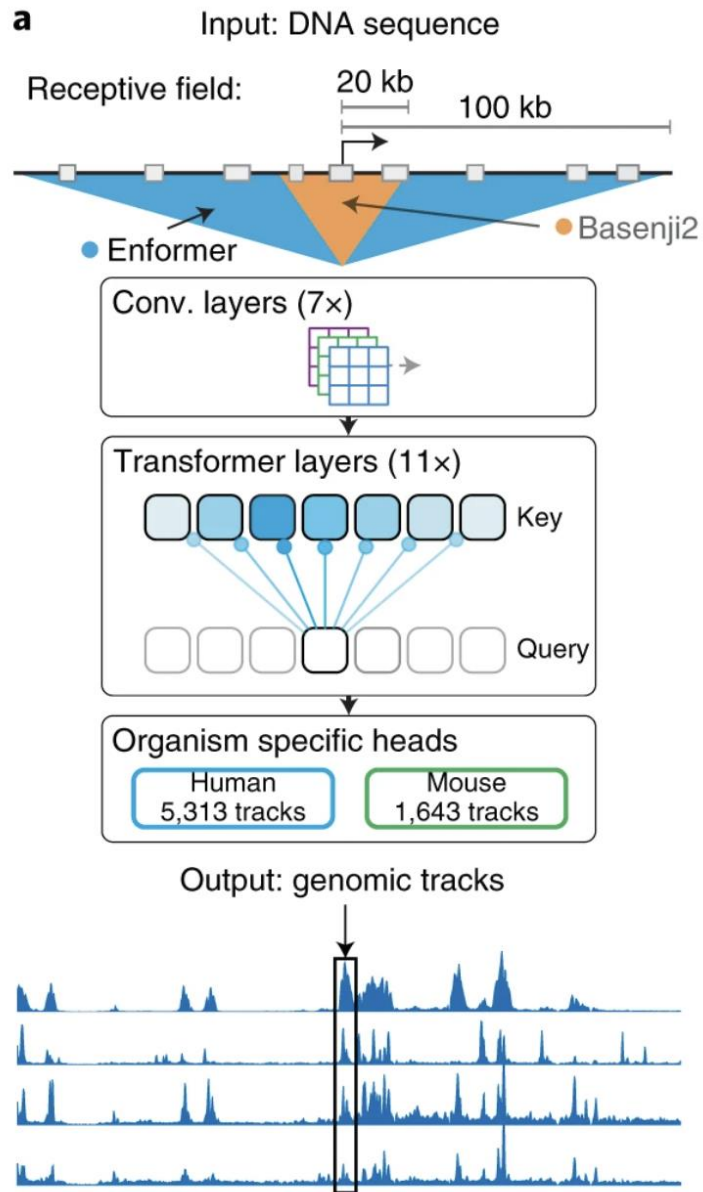
Effective gene expression prediction from sequence by integrating long-range interactions

[Žiga Avsec](#) , [Vikram Agarwal](#), [Daniel Visentin](#), [Joseph R. Ledsam](#), [Agnieszka Grabska-Barwinska](#), [Kyle R. Taylor](#), [Yannis Assael](#), [John Jumper](#), [Pushmeet Kohli](#)  & [David R. Kelley](#) 

[Nature Methods](#) **18**, 1196–1203 (2021) | [Cite this article](#)

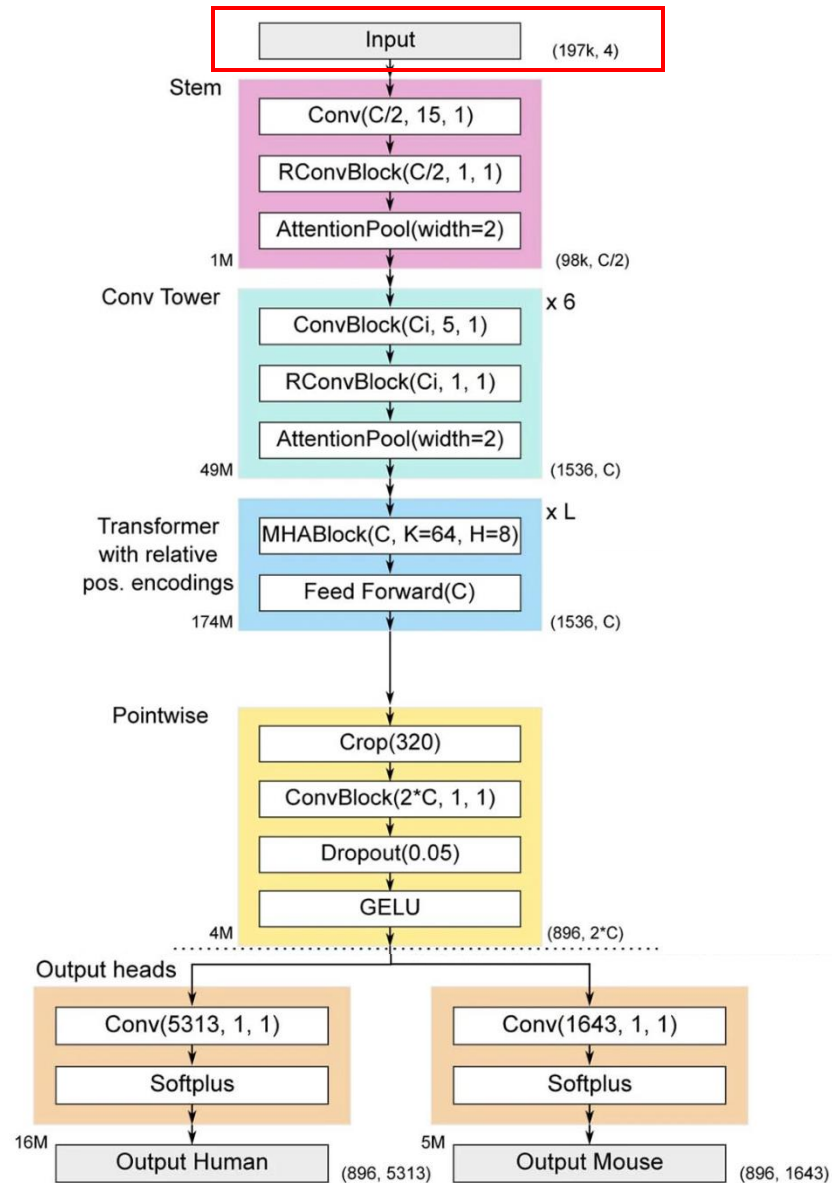
306k Accesses | **1001** Citations | **380** Altmetric | [Metrics](#)

- The most widely used genomic Large Language Model (LLM)
- Developed in the first wave of genomic LLMs
- Has since defined the field



- Trained on **197kb** DNA sequences from the human and mouse **reference** genomes
- Each sequence has thousands of corresponding labels; **supervised** model
 - **ATAC-seq**
 - Is this sequence available to be transcribed?
 - **DNase-seq**
 - Same as ATAC-seq, but different measurement
 - **ChIP-seq**
 - Are proteins binding to this sequence?
 - **CAGE**
 - How much of this sequence is transcribed into RNA?
- Model predicts **all labels** at **each position**

Enformer (C=1536, L=11)

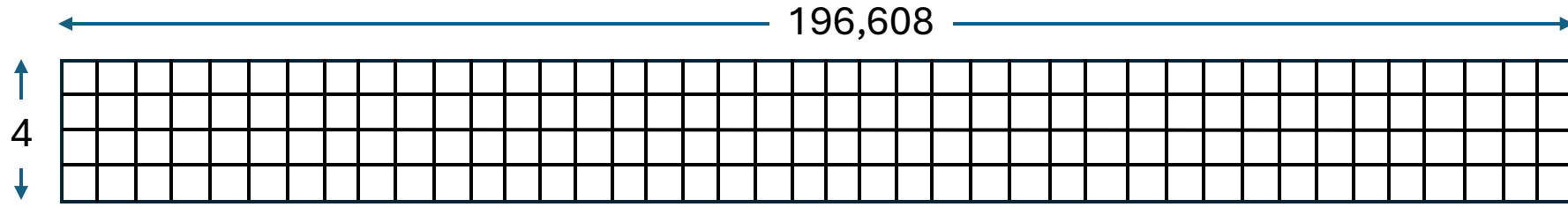


Input is 197k one-hot encoded DNA bases

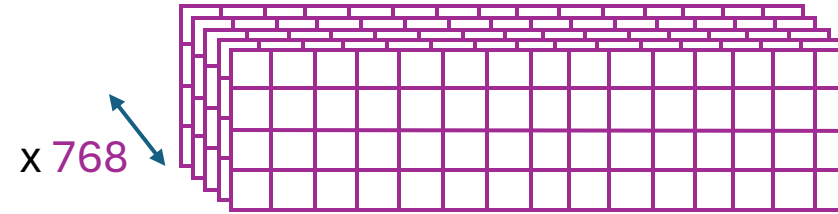
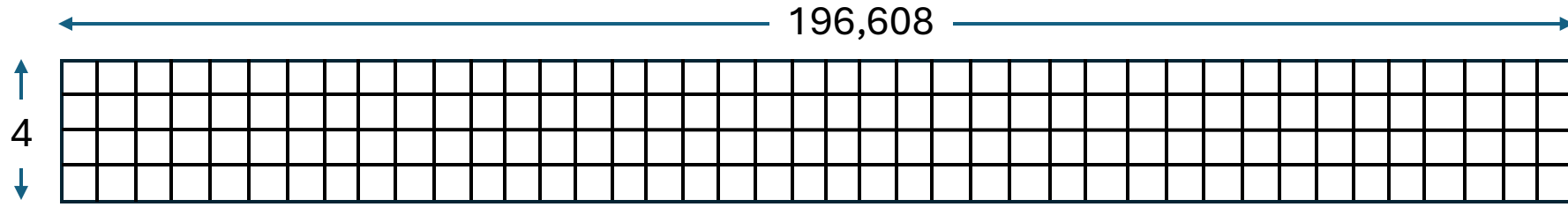
GATTACA

[0 0 1 0], [1 0 0 0], [0 0 0 1], [0 0 0 1], [1 0 0 0], [0 1 0 0], [1 0 0 0]

Input:

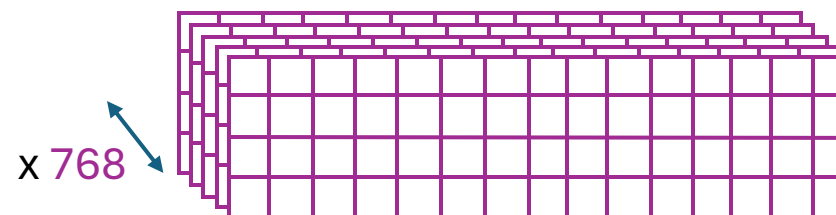
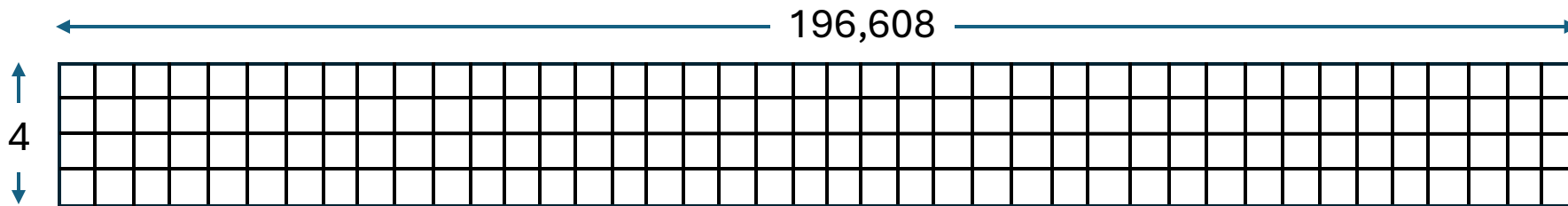


Input:

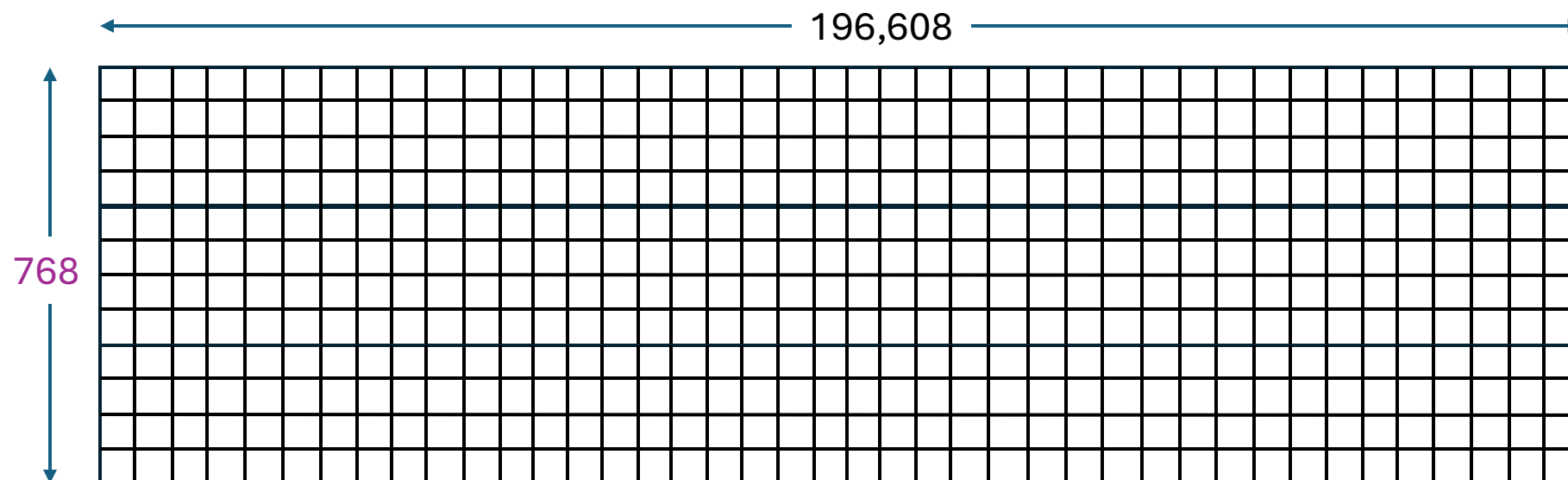


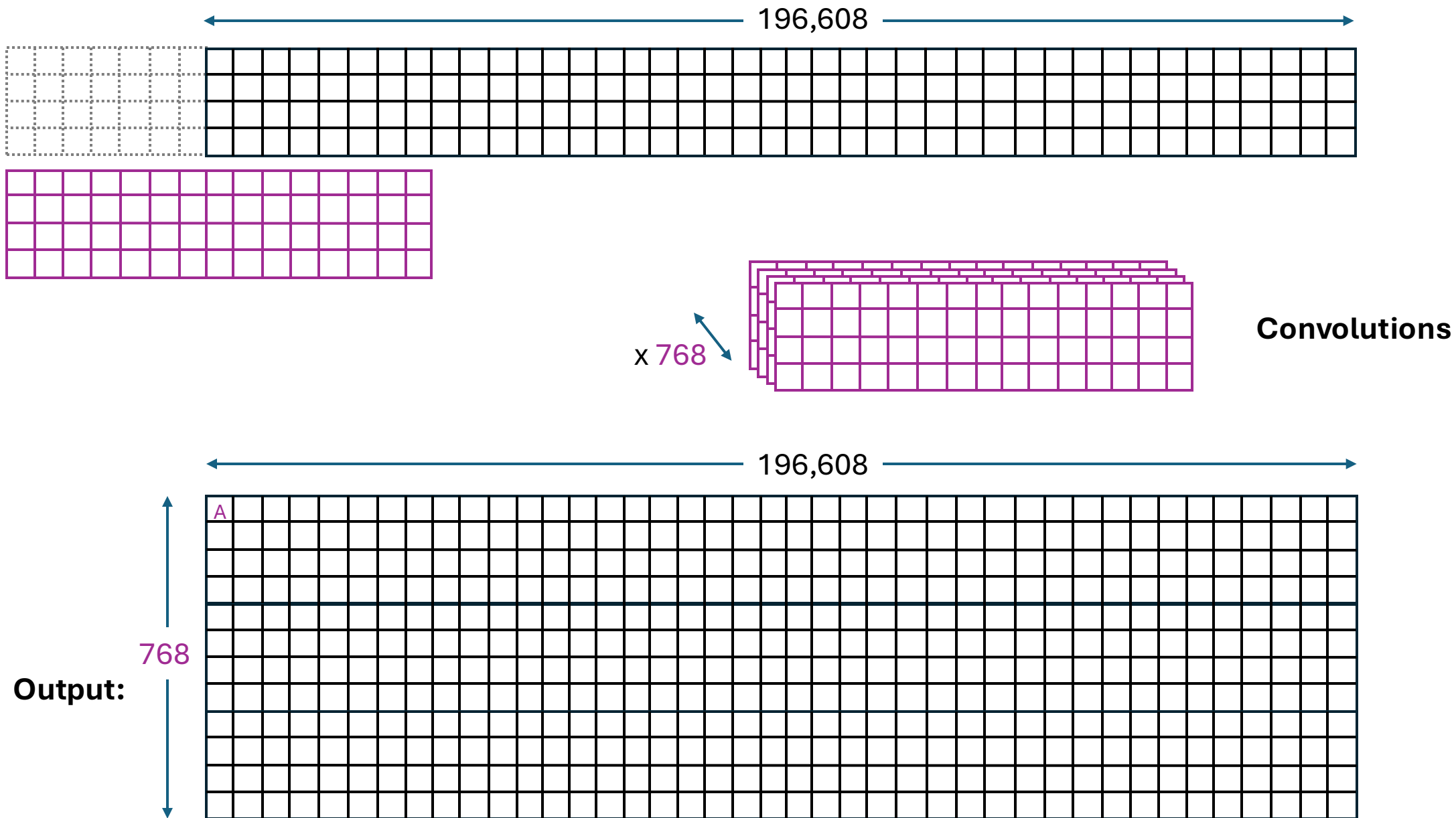
Convolutions

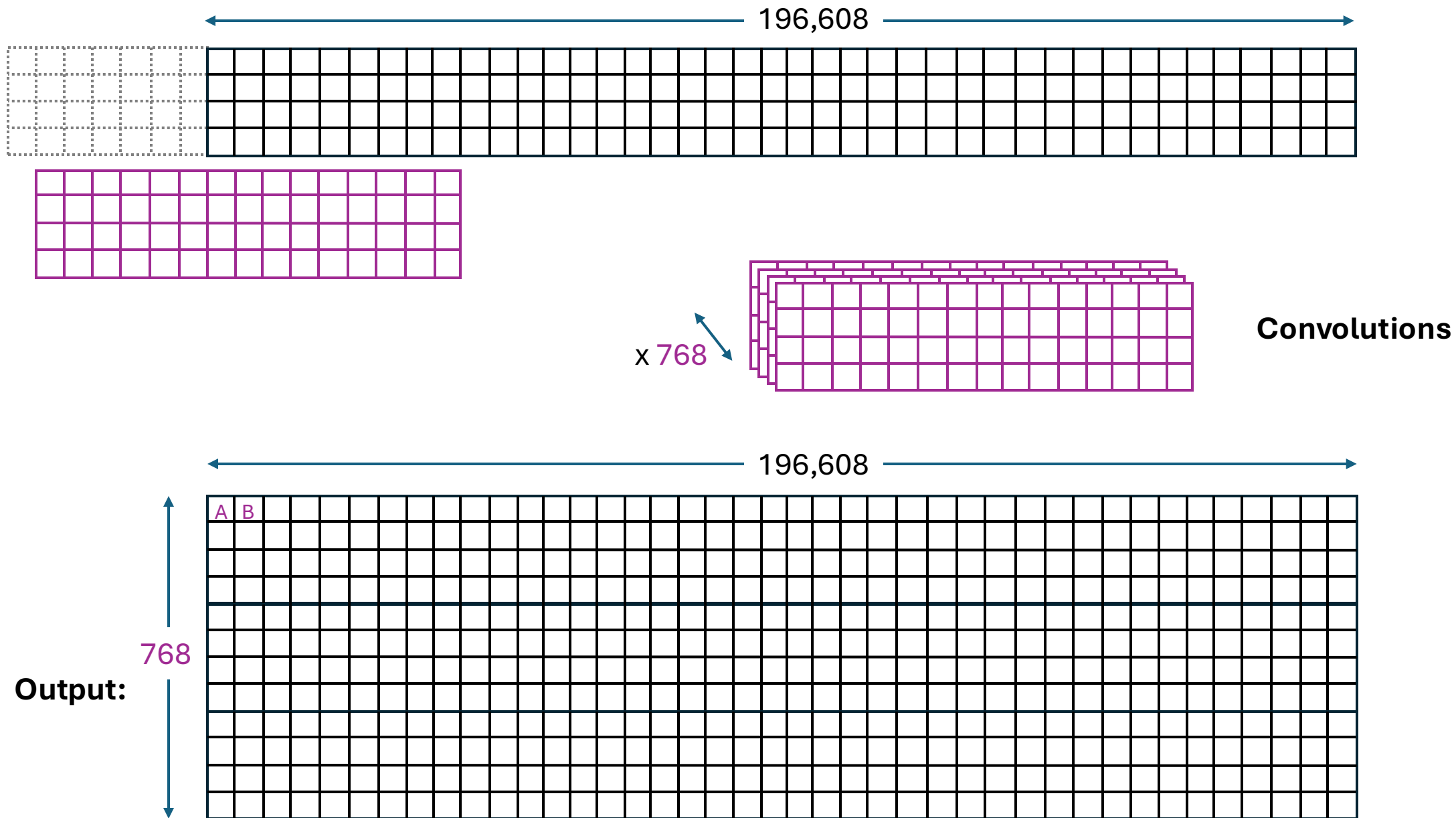
Input:

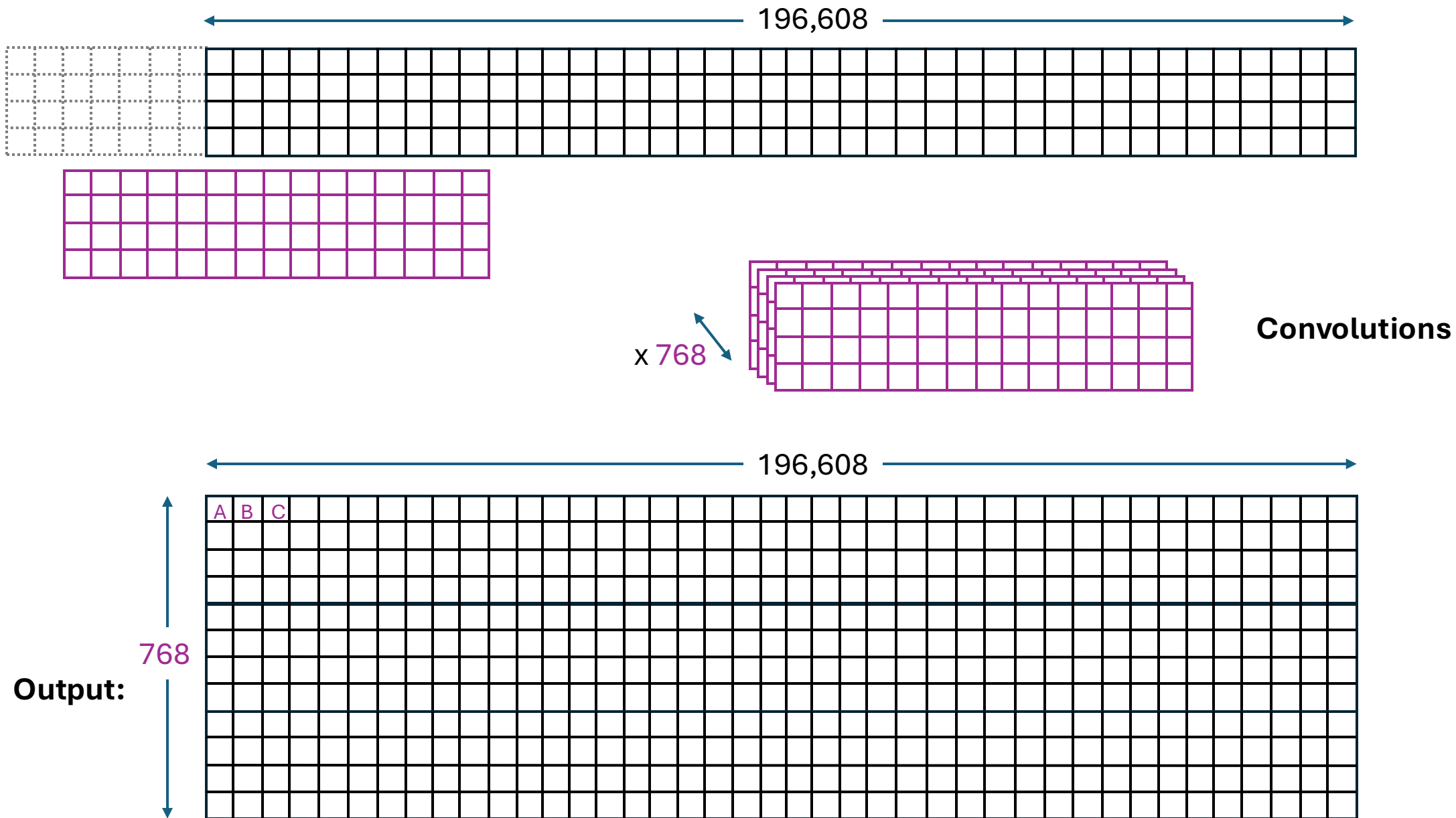


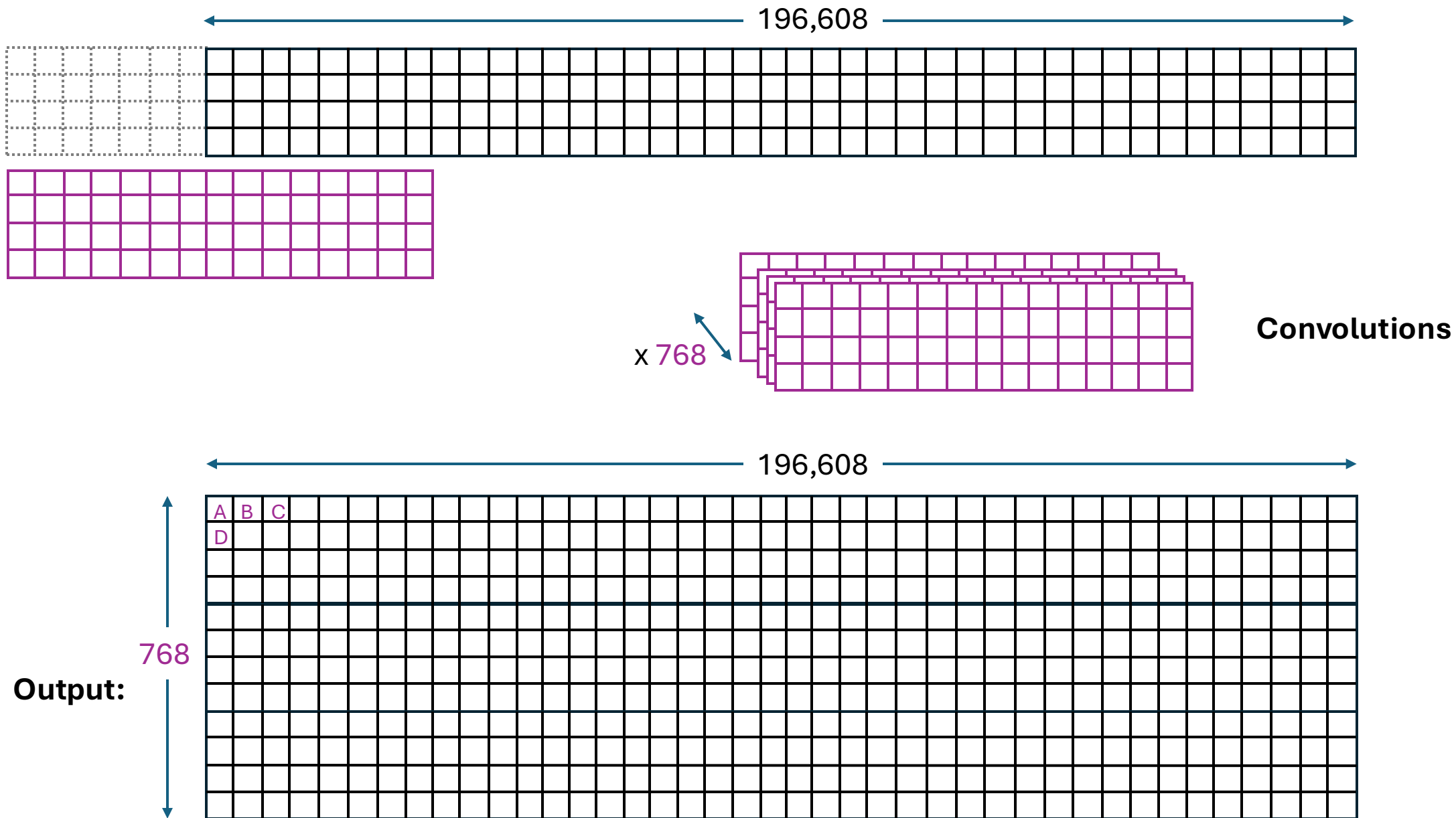
Output:

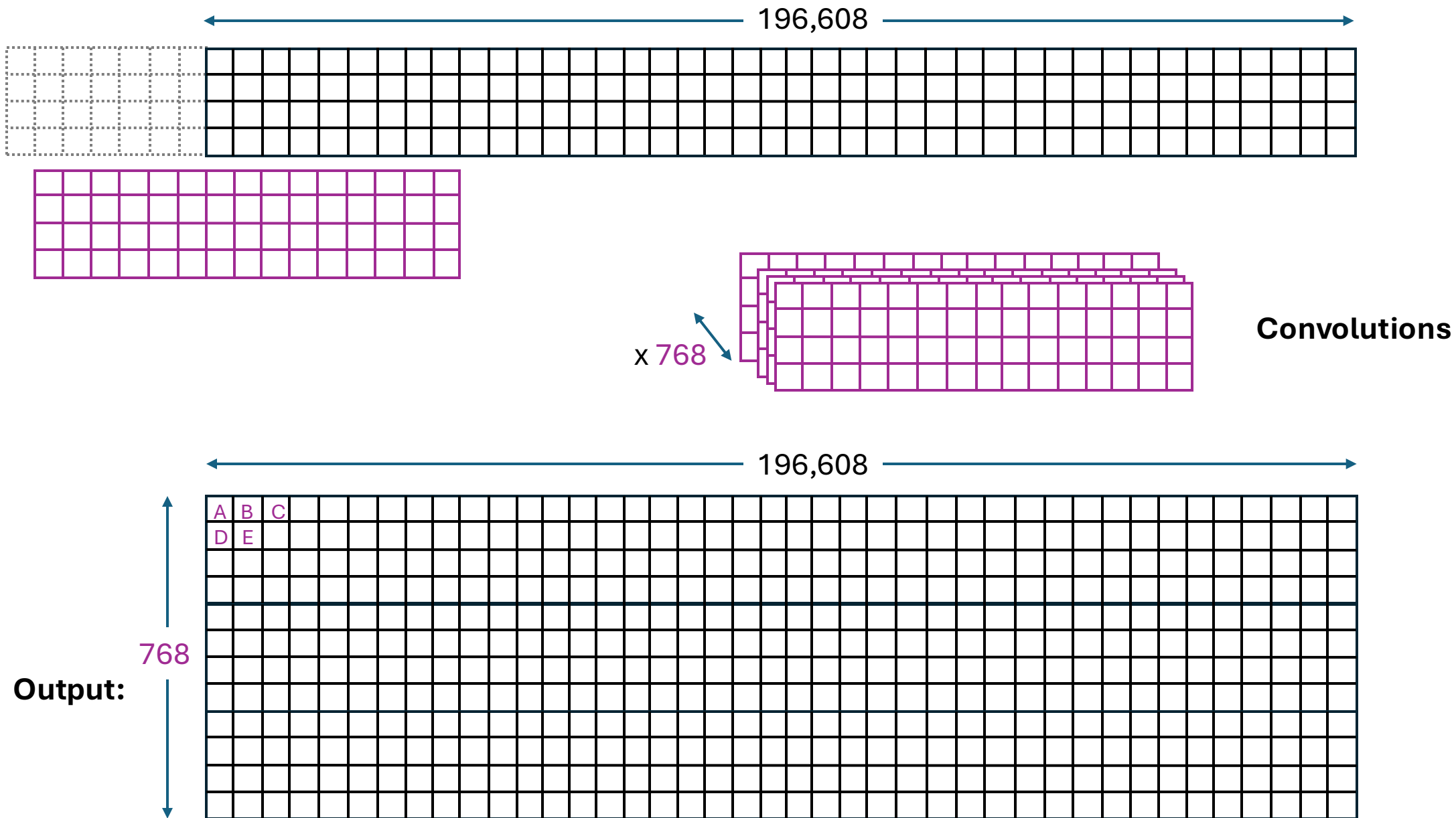


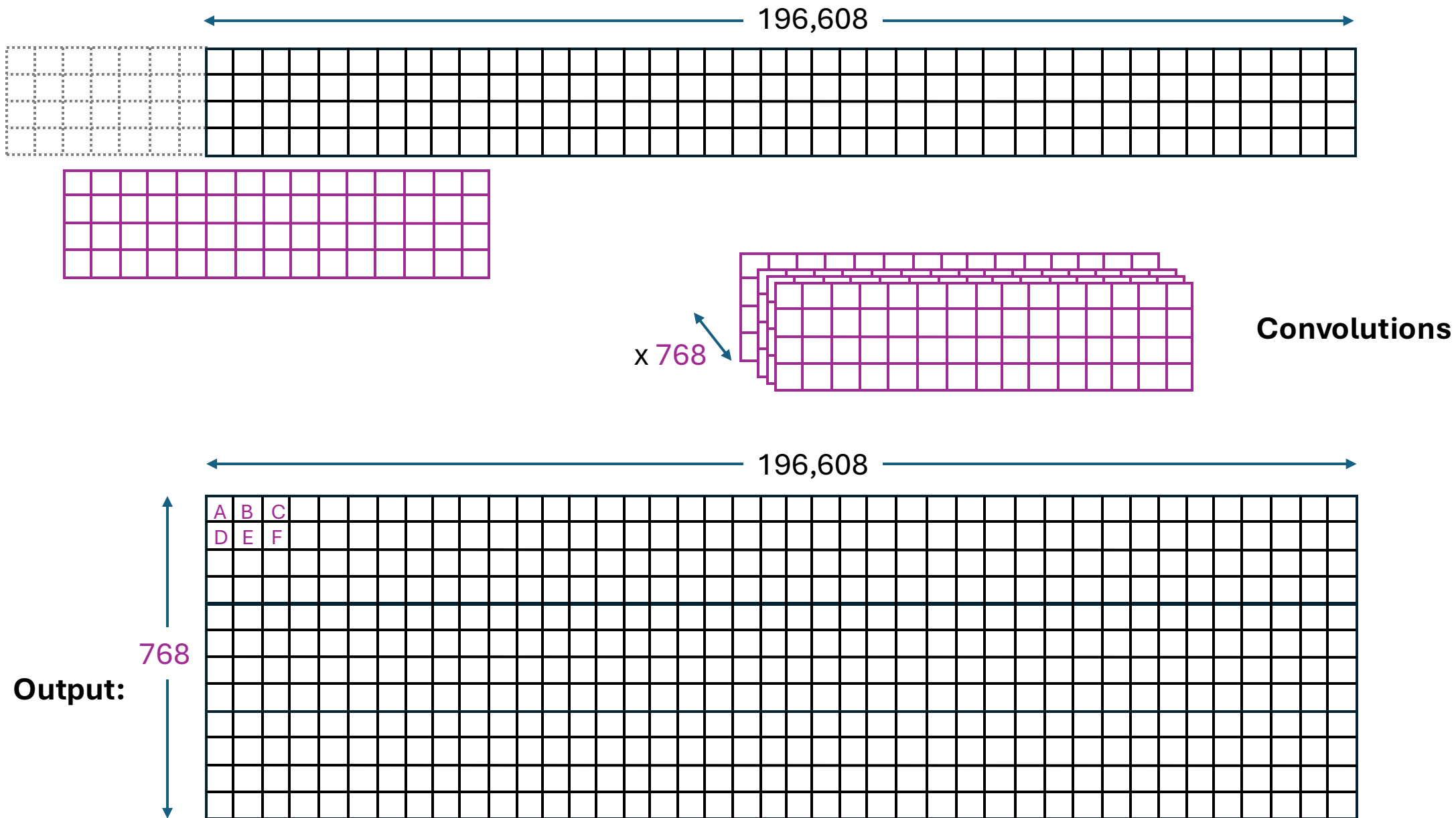




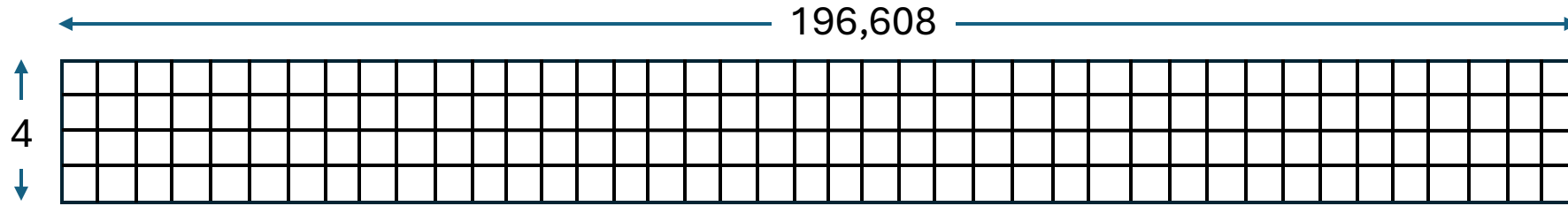






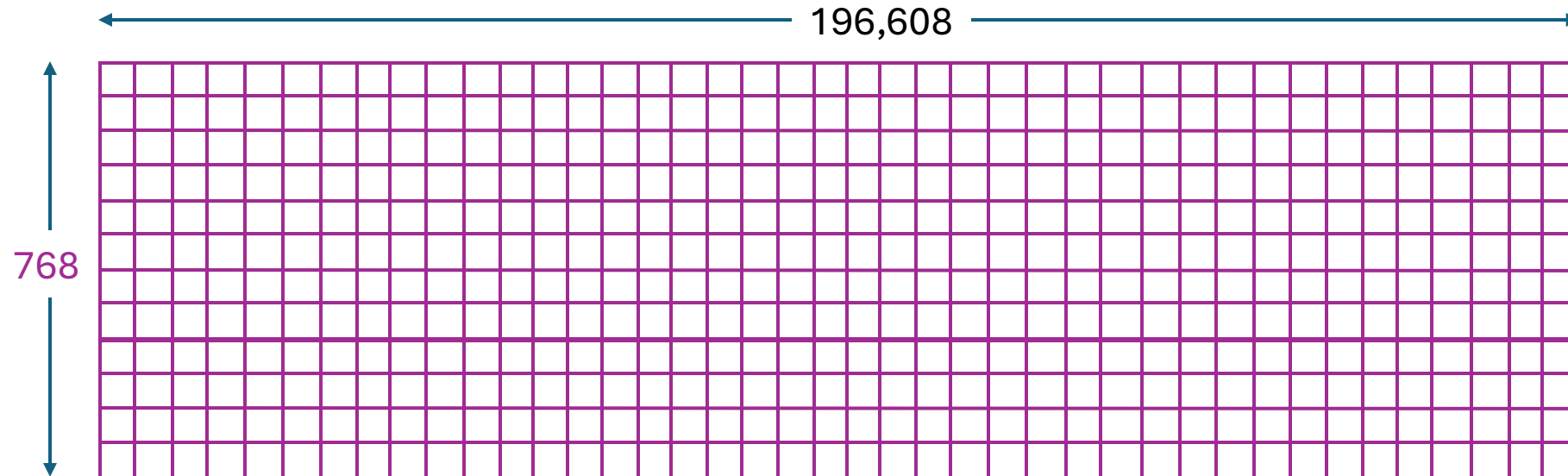


Input:

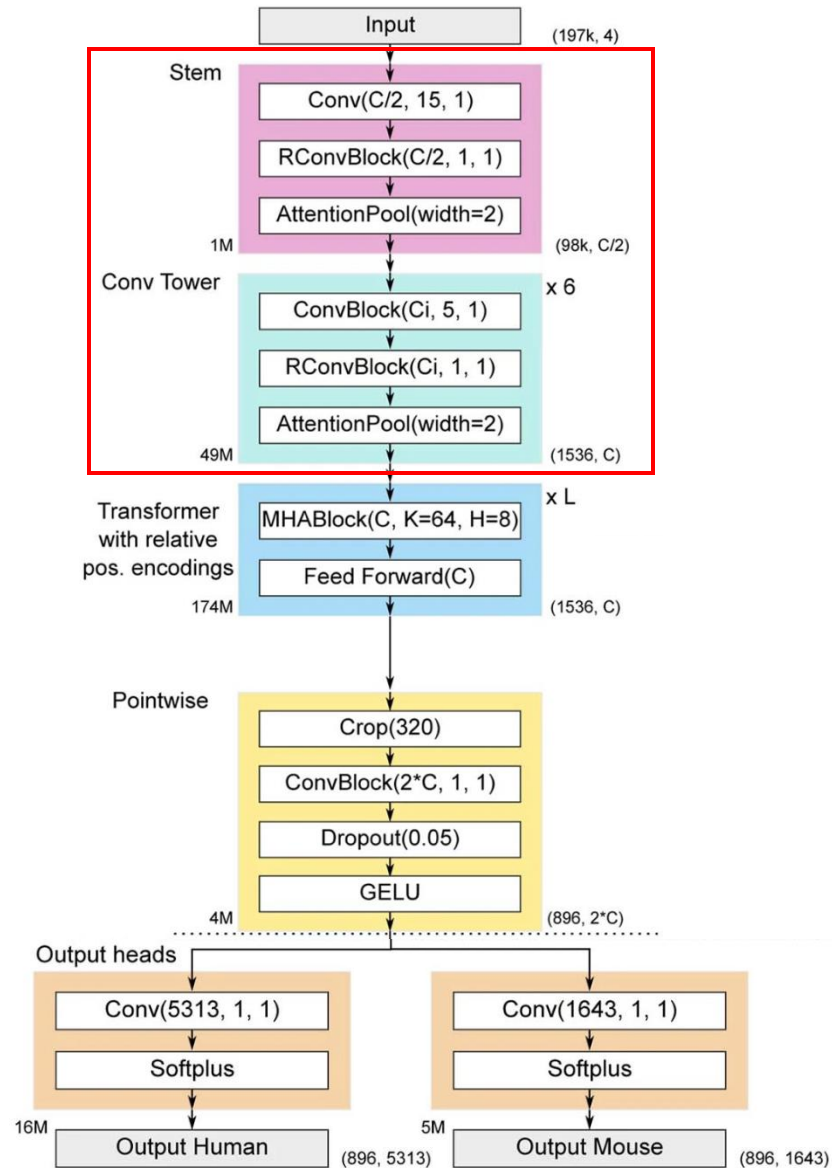


Convolutions **expand** data from **4** to **768 channels** while preserving length
This step is critical to giving the model enough **information** to work with

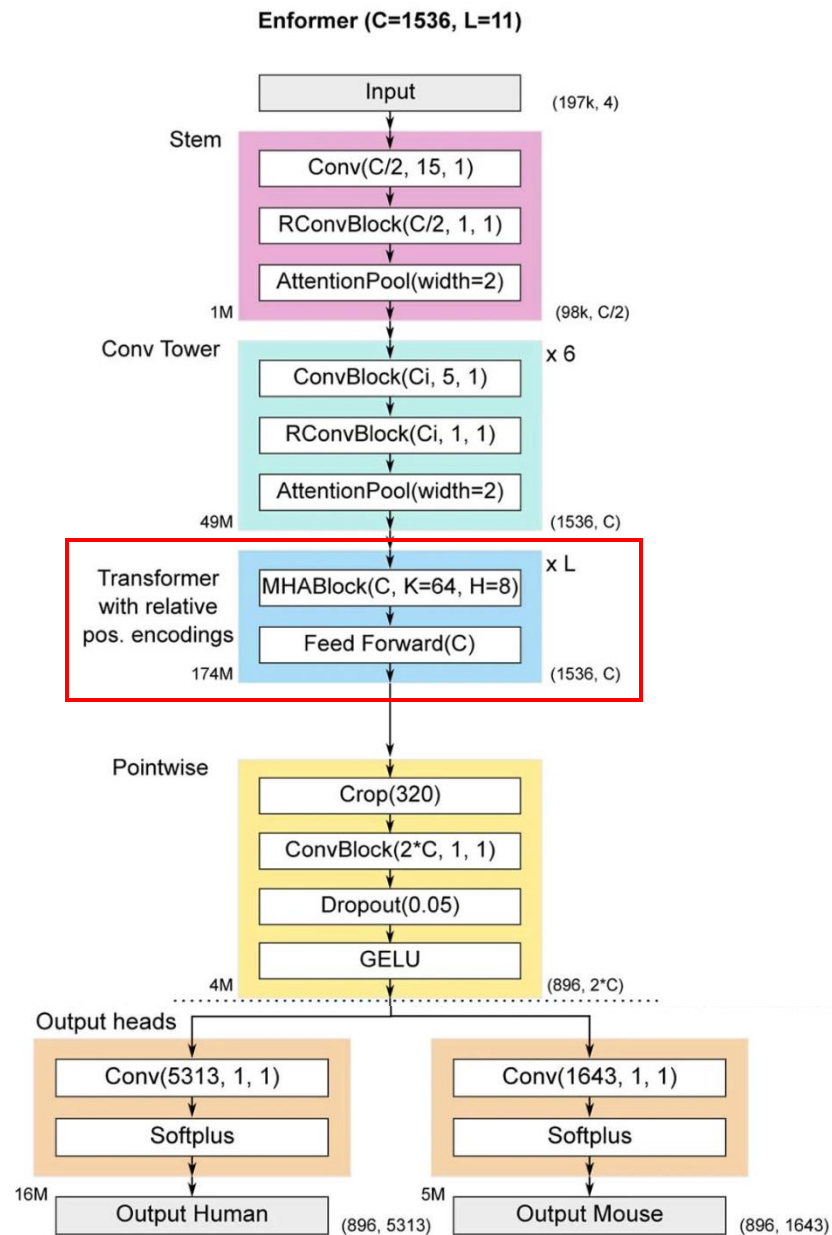
Output:



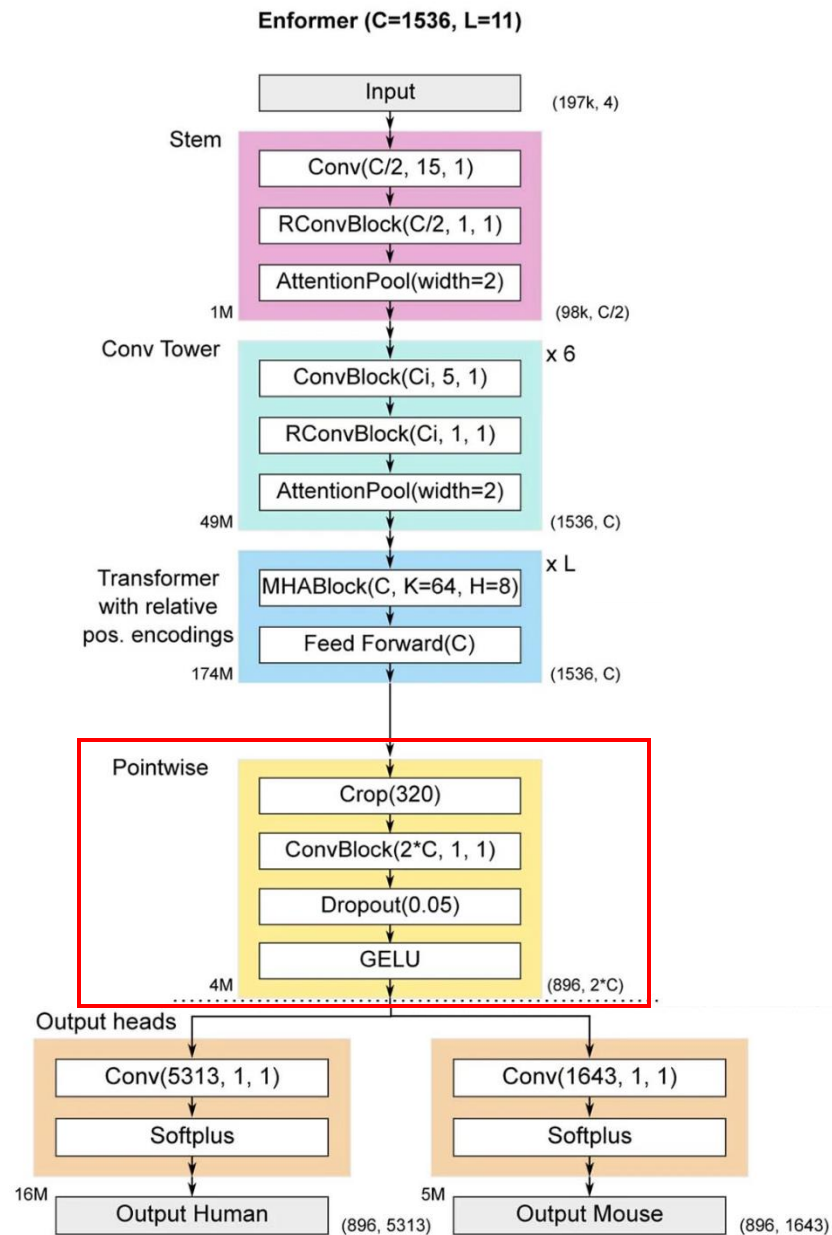
Enformer (C=1536, L=11)



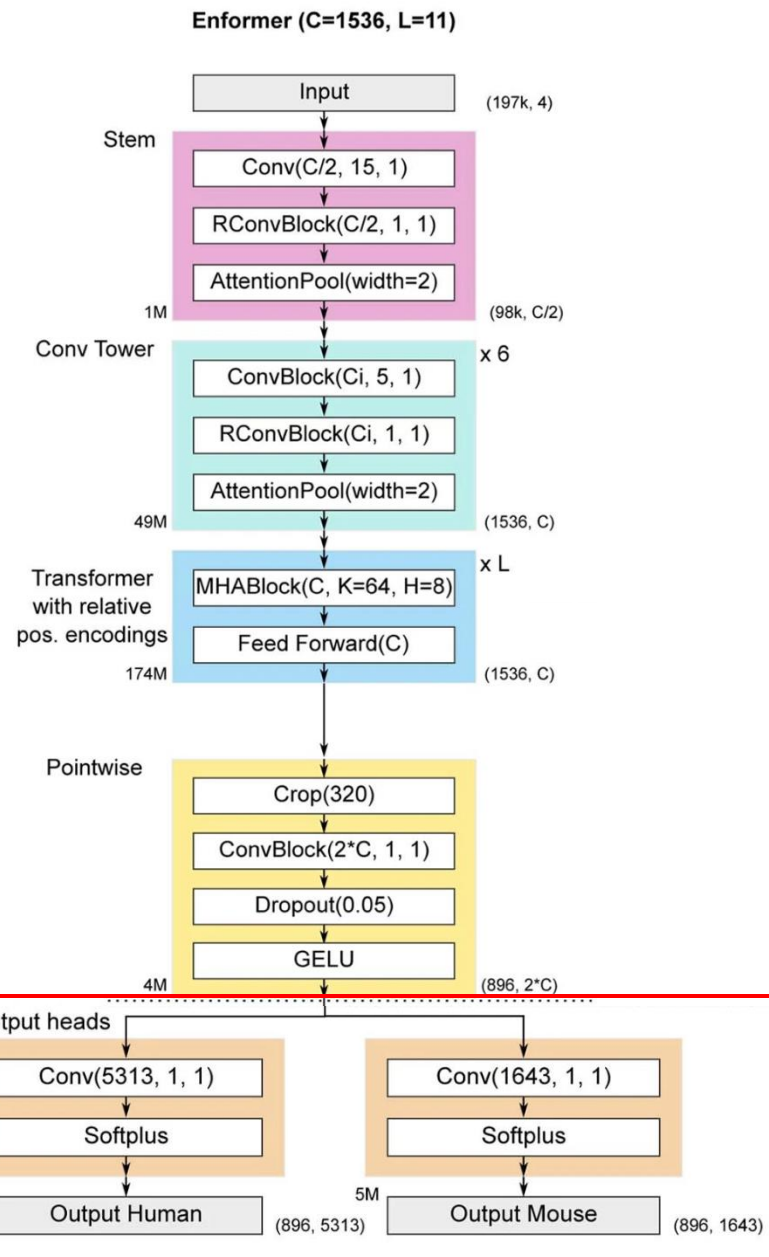
Increase the dimensionality of the data (4 -> 1536 channels)
Decrease the number of data points (197k -> 1536)



Add **context** to data, like combining word vectors into sentence vectors
The power of transformers



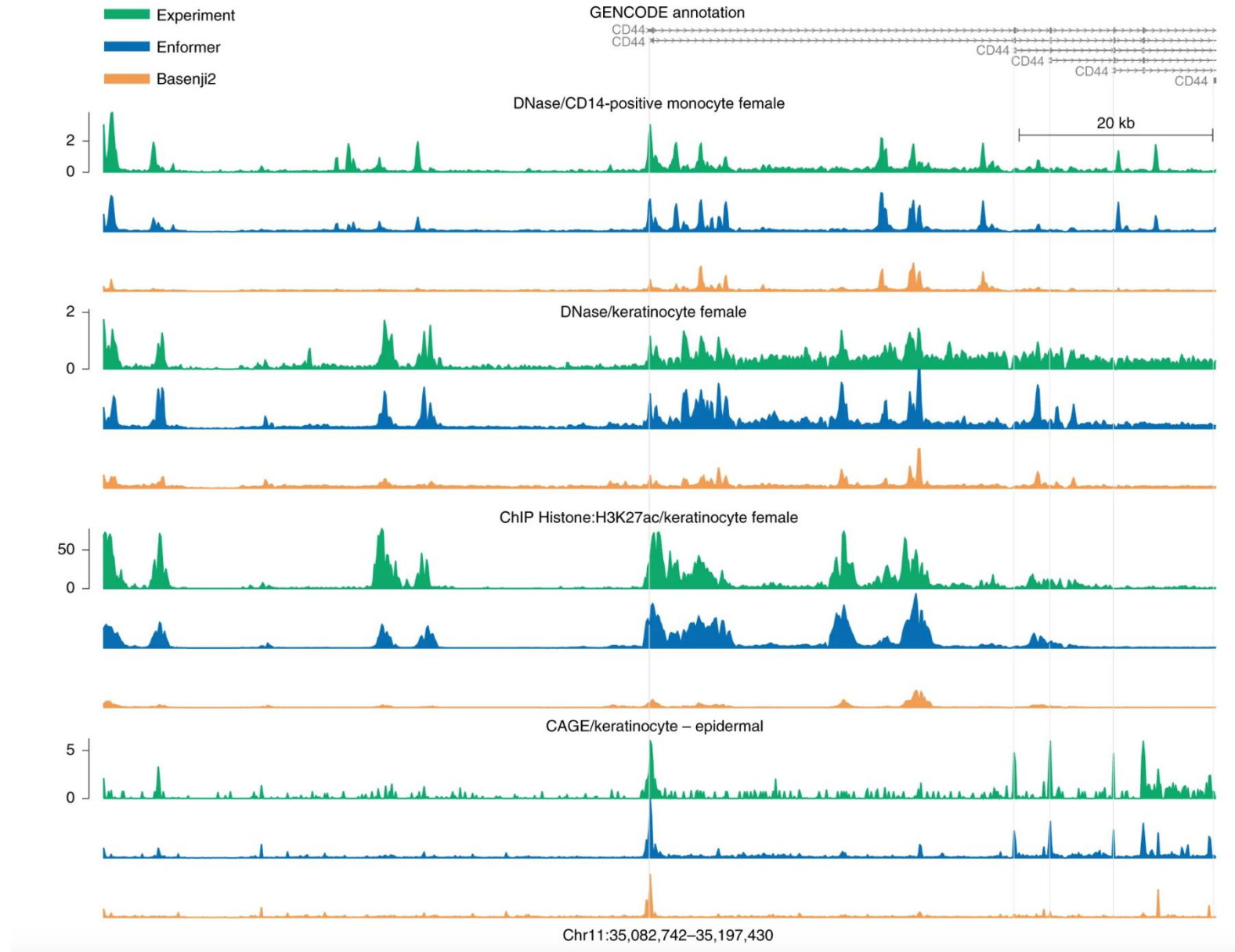
Fully connected network
Learn **patterns** in data



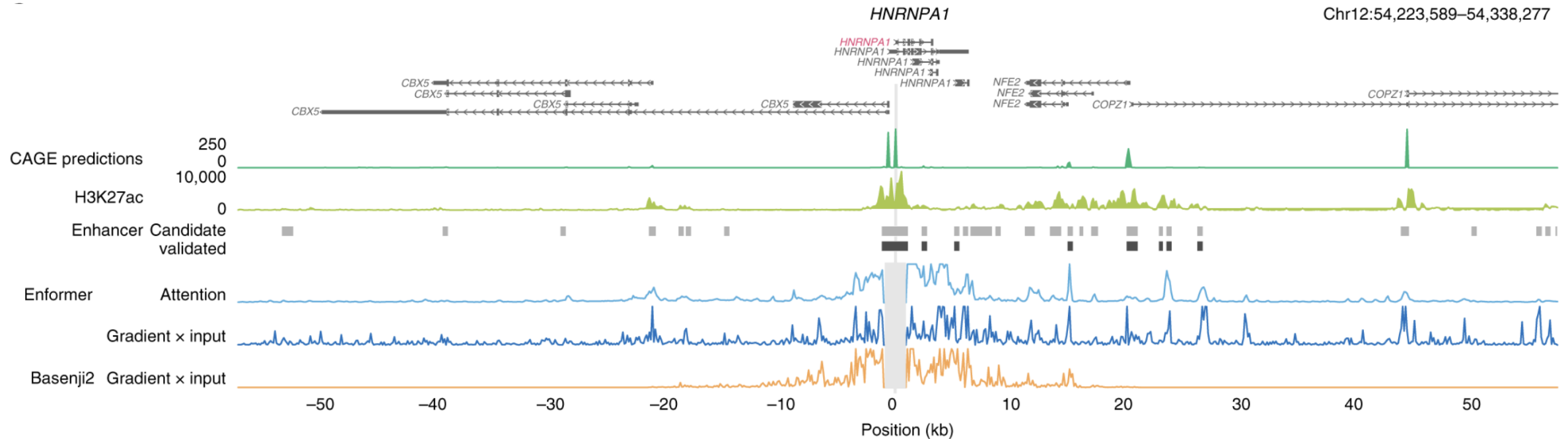
Emit **predictions** across all labels

What does Enformer do?

- Green: test sample **experimental** data
- Blue: Enformer's **predictions**
- Training enables it to **generalize** to unseen samples

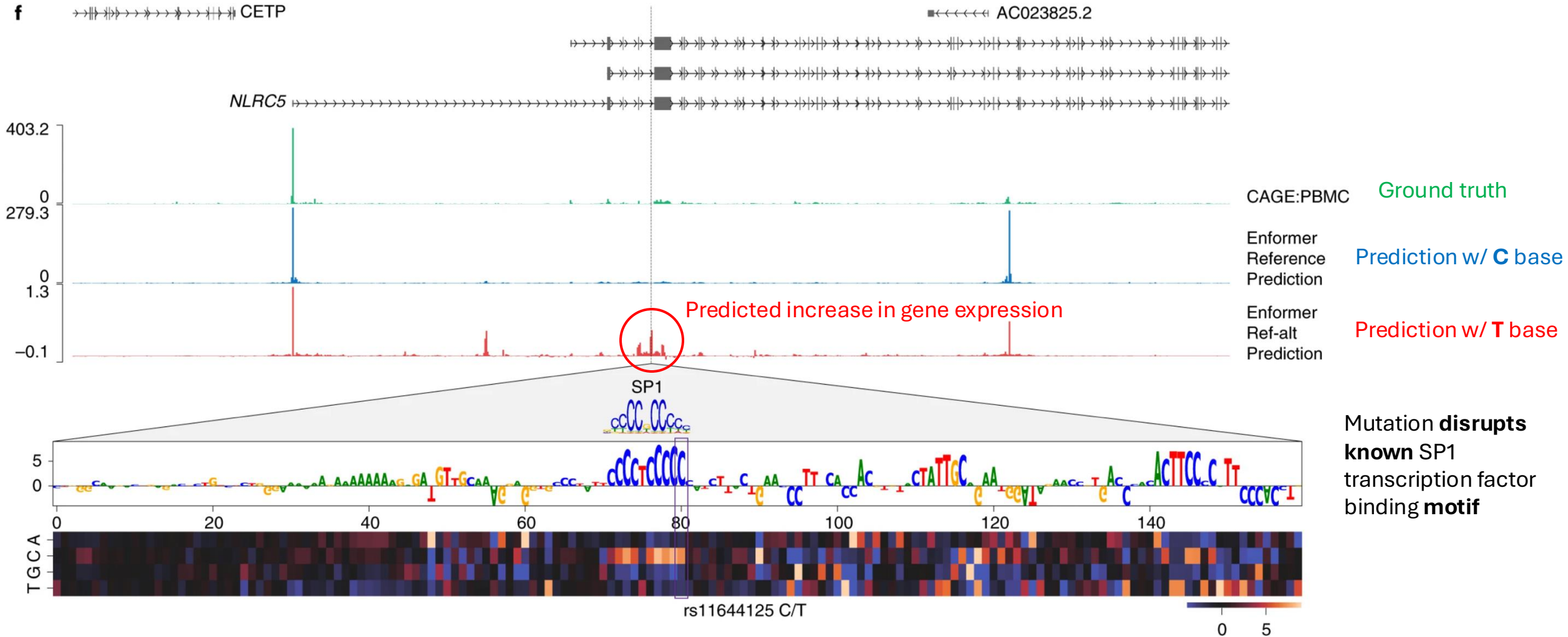


Interpreting Enformer's predictions



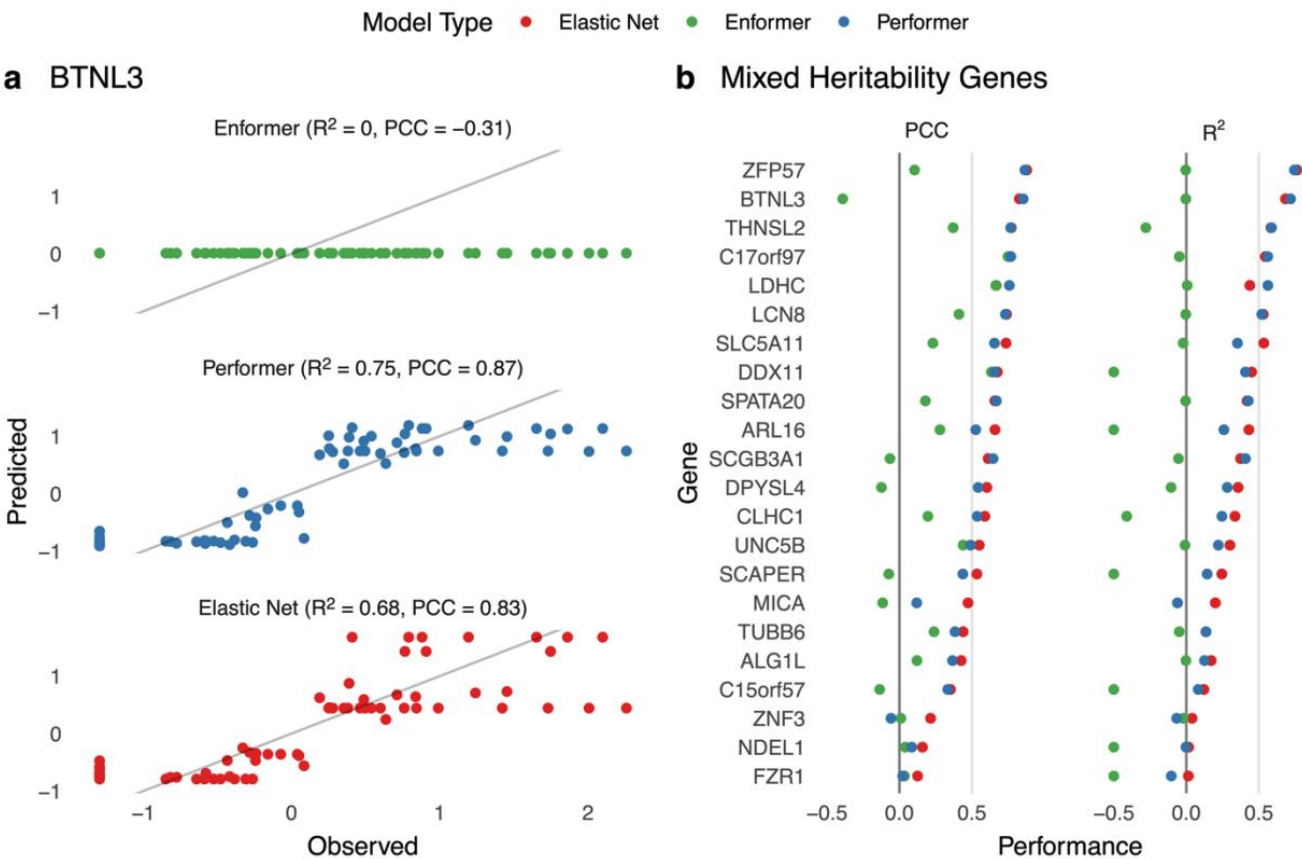
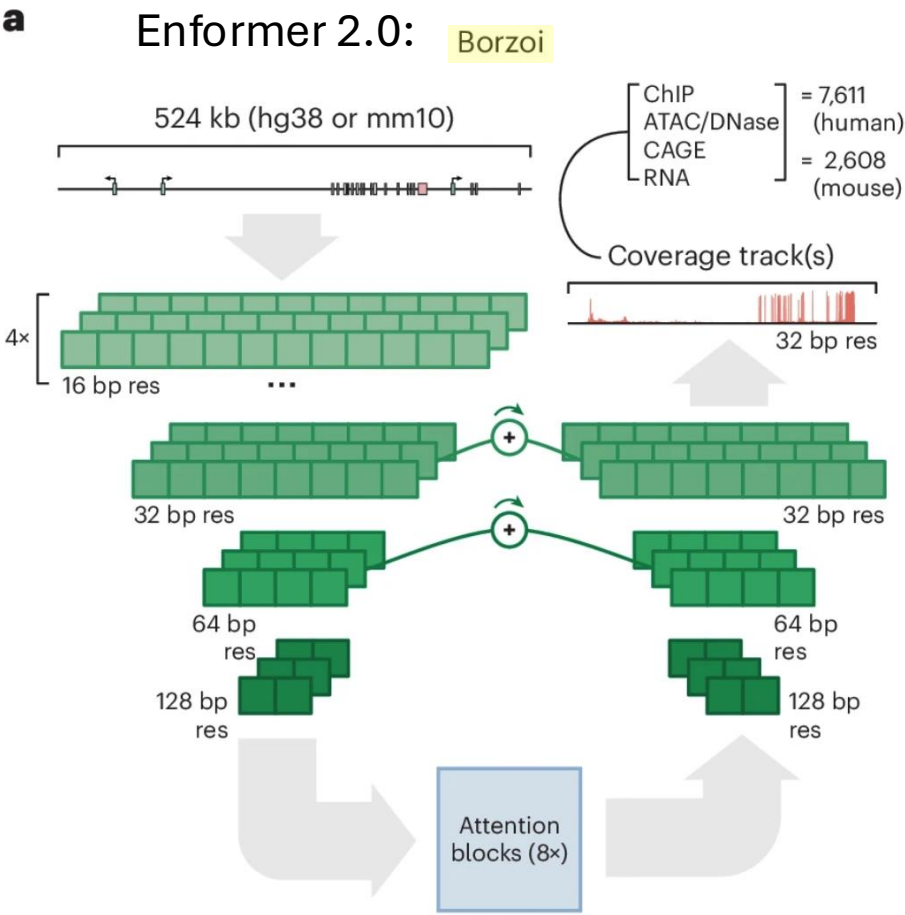
- Dark green: Enformer's CAGE **prediction** of transcript abundance for *HNRNPA1*
- Light green and gray: **known** enhancer positions for *HNRNPA1*
- Blues: **attention** and **gradient** values at each position of the input
- Locations of high attention and gradients **correspond** with known enhancers; looking in the **right places**

Understanding variants with Enformer



Enformer's influence

Several models fine-tuning Enformer for variant effect prediction, such as **Performer**



Genome modeling and design across all domains of life with Evo 2

Evo2

- The largest genomic LLM created to date
- Published February 2025
- Very divisive in the field

Garyk Brixi^{*,1,2,3}, Matthew G. Durrant^{*,1,2}, Jerome Ku^{*,1,2}, Michael Poli^{*,2,3,5},
Greg Brockman^{**,2,6,8}, Daniel Chang^{**,1,2,3}, Gabriel A. Gonzalez^{**,1,2}, Samuel H. King^{**,1,2,3},
David B. Li^{**,1,2,3}, Aditi T. Merchant^{**,1,2,3}, Mohsen Naghipourfar^{**,1,2,7}, Eric Nguyen^{**,2,3},
Chiara Ricci-Tam^{**,1,2}, David W. Romero^{**,2,4}, Gwanggyu Sun^{**,1,2}, Ali Taghibakshi^{**,2,4},
Anton Vorontsov^{**,2,4}, Brandon Yang^{**,2,6}, Myra Deng⁸, Liv Gorton⁸, Nam Nguyen⁸,
Nicholas K. Wang⁸, Etowah Adams⁹, Stephen A. Baccus³, Steven Dillmann³,
Stefano Ermon³, Daniel Guo^{1,3}, Rajesh Ilango¹, Ken Janik⁴, Amy X. Lu⁷, Reshma Mehta⁶,
Mohammad R.K. Mofrad⁷, Madelena Y. Ng³, Jaspreet Pannu³, Christopher Ré³,
Jonathan C. Schmok¹, John St. John⁴, Jeremy Sullivan¹, Kevin Zhu⁷, Greg Zynda⁴,
Daniel Balsam^{8,10}, Patrick Collison^{1,10}, Anthony B. Costa^{4,10}, Tina Hernandez-
Boussard^{3,10}, Eric Ho^{8,10}, Ming-Yu Liu^{4,10}, Thomas McGrath^{8,10},
Kimberly Powell^{4,10}, Dave P. Burke^{‡,1,2,10}, Hani Goodarzi^{‡,1,2,10,11},
Patrick D. Hsu^{‡,†,1,2,7,10}, Brian L. Hie^{‡,†,1,2,3,10}

¹Arc Institute; ²Core Contributor, Evo 2 Team; ³Stanford University; ⁴NVIDIA;
⁵Liquid AI; ⁶Independent Researcher; ⁷University of California, Berkeley;
⁸Goodfire; ⁹Columbia University; ¹⁰Senior Contributor, Evo 2 Team;
¹¹University of California, San Francisco

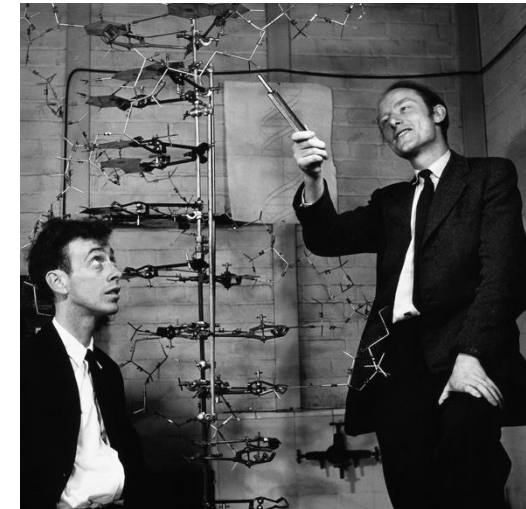
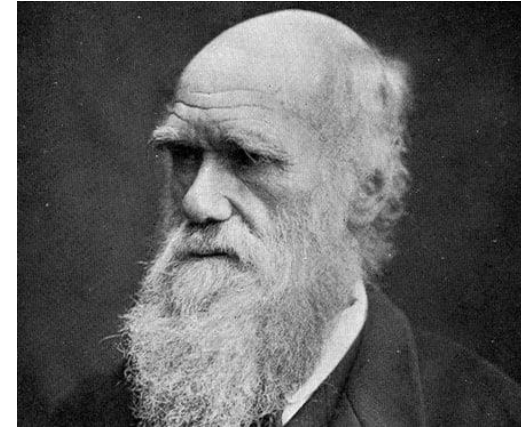
Extreme training requirements

- “Evo 2 was trained for **several months** on the NVIDIA DGX Cloud AI platform via AWS, utilizing over **2,000 NVIDIA H100 GPUs** and bolstered by **collaboration with NVIDIA** researchers and engineers.” - Arc Institute blog post
- Roughly **\$10M** training



Braggadocious framing

- “Biological research scales from molecules to systems to organisms, seeking to understand and design functional components across all domains of life (**Darwin**, 1859; **Mendel**, 1866; Dobzhansky, 1951) ... All domains of life express complex functions from DNA sequences (**Watson and Crick**, 1953; Nirenberg and Matthaei, 1961), yet genomic content and length vary dramatically across organisms.”



Difficulty of use

- Evo2-40B weights take up **80GB** in GPU RAM
- H100 node required to compile the code and run inference
- Can take **hours** of inference time to process a **single sample**



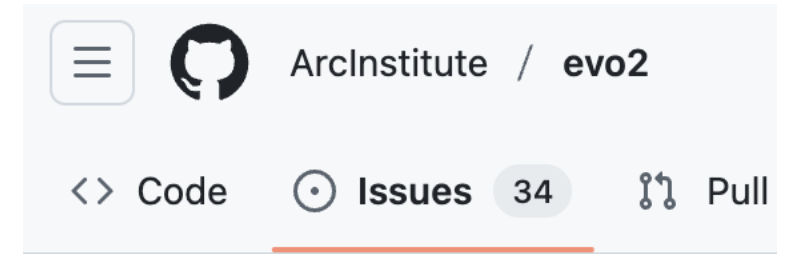
Nvidia H100 NVL GPU Computing Processor

\$29,500.00

Pre-owned

 eBay & more

4.0 ★★★★★ (1)

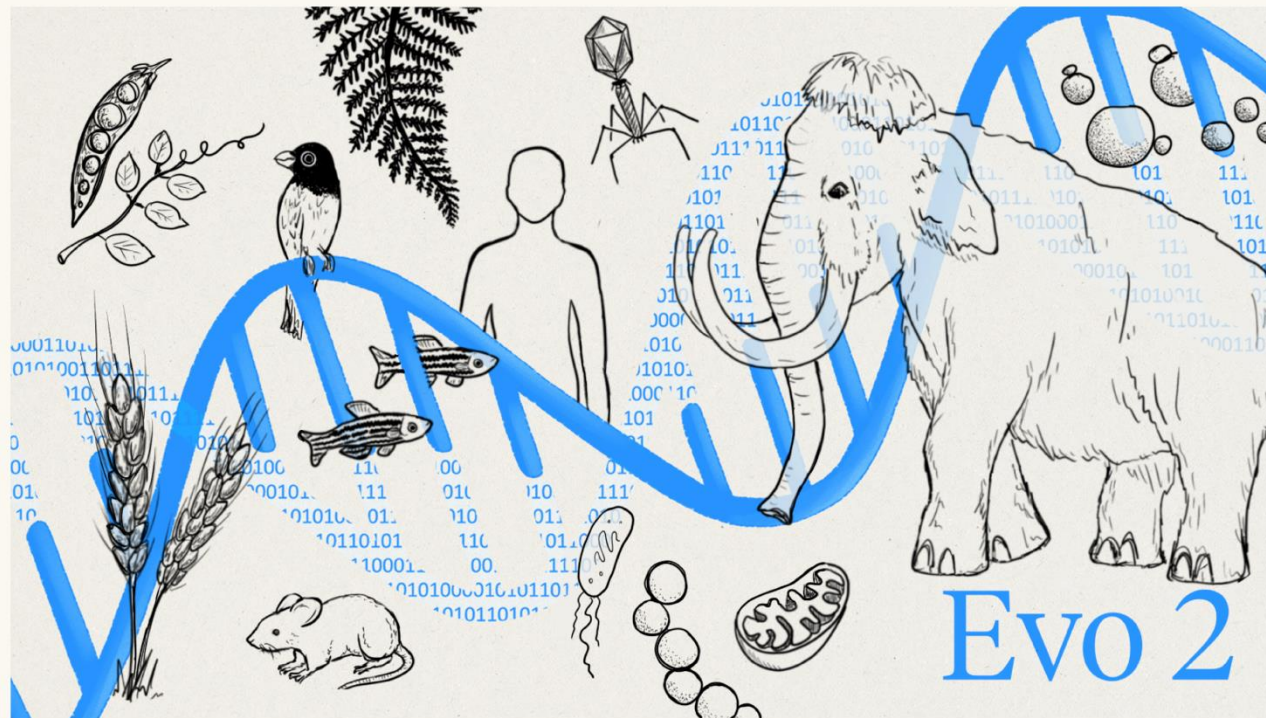


is:issue state:open

Open 34 Closed 100

What makes Evo2 so impressive?

Arc Institute develops the largest AI model for biology to date in collaboration with NVIDIA, bringing together Stanford University, UC Berkeley, and UC San Francisco researchers



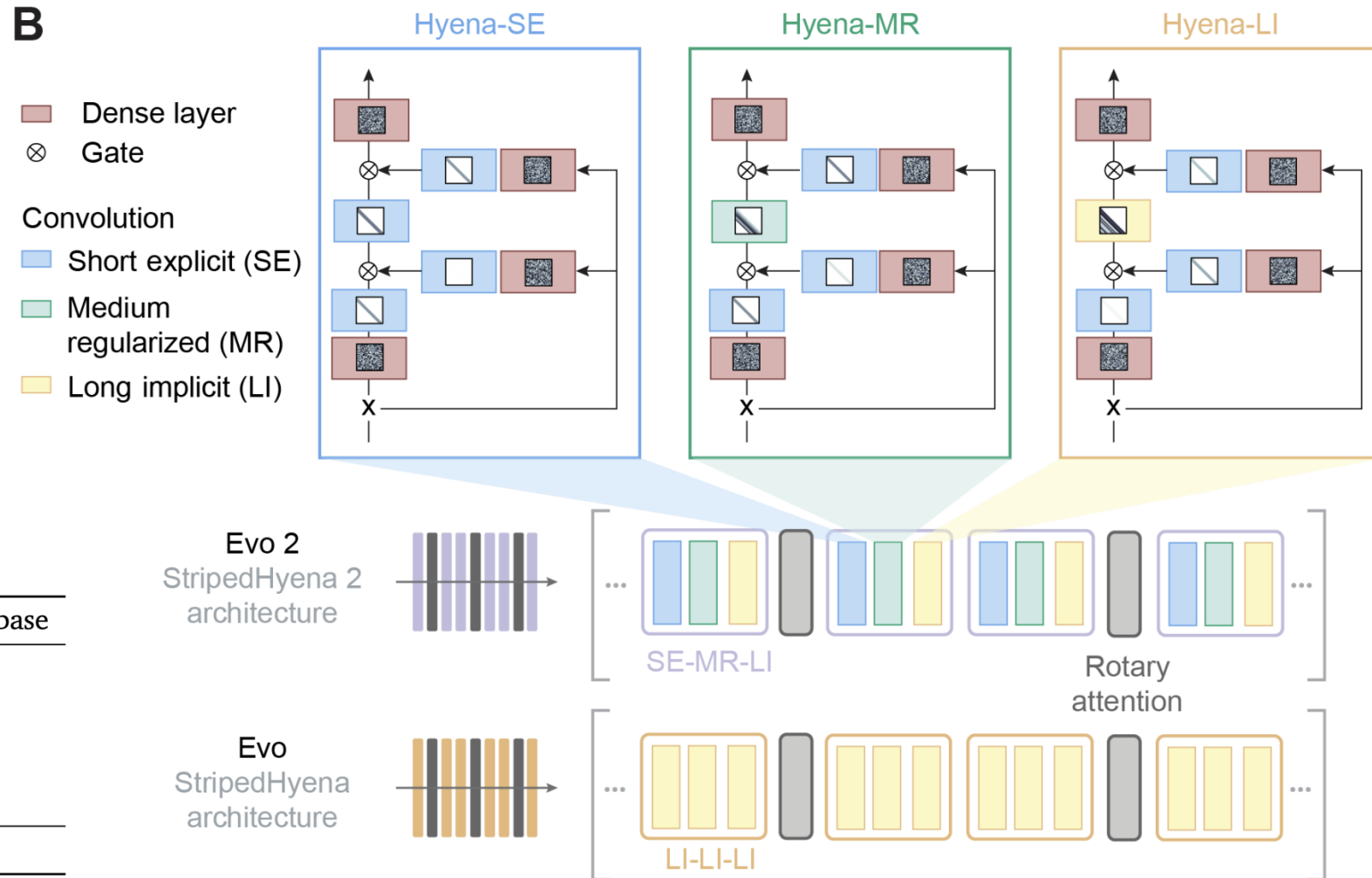
Evo 2 is trained on over 9.3 trillion tokens—in this case, nucleotides—from over 128 thousand genomes across the three domains of life, making it similar in scale to the most powerful generative AI large language models.

Model architecture

- StripedHyena 2
- Intentionally **patterned** series of **convolutions** and **attention** layers
- Mix of 7, 128, and full-length convolutions

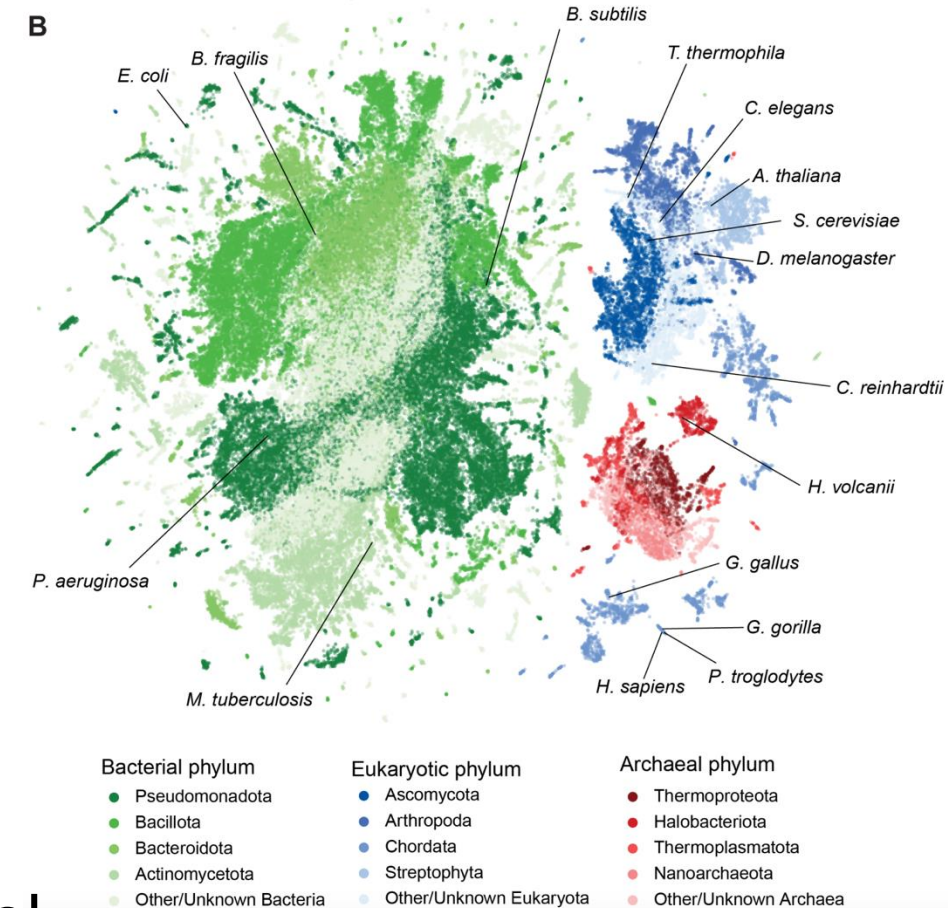
	Evo 2 40B	Evo 2 7B	Evo 2 1B base
Parameters	40.3B	6.5B	1.1B
Total Layers	50	32	25
Hidden Size	8,192	4,096	1,920
FFN Size	22,528	11,264	5,120
Num Heads	64	32	15
Total Tokens	9.3T	2.4T	1T

- **Convolutions** find **features**
- **Attention** gives features **context**
- **Early** layer features are **simple**, like kmer motifs
- **Late** layer features are **complex**, like gene-specific CREs

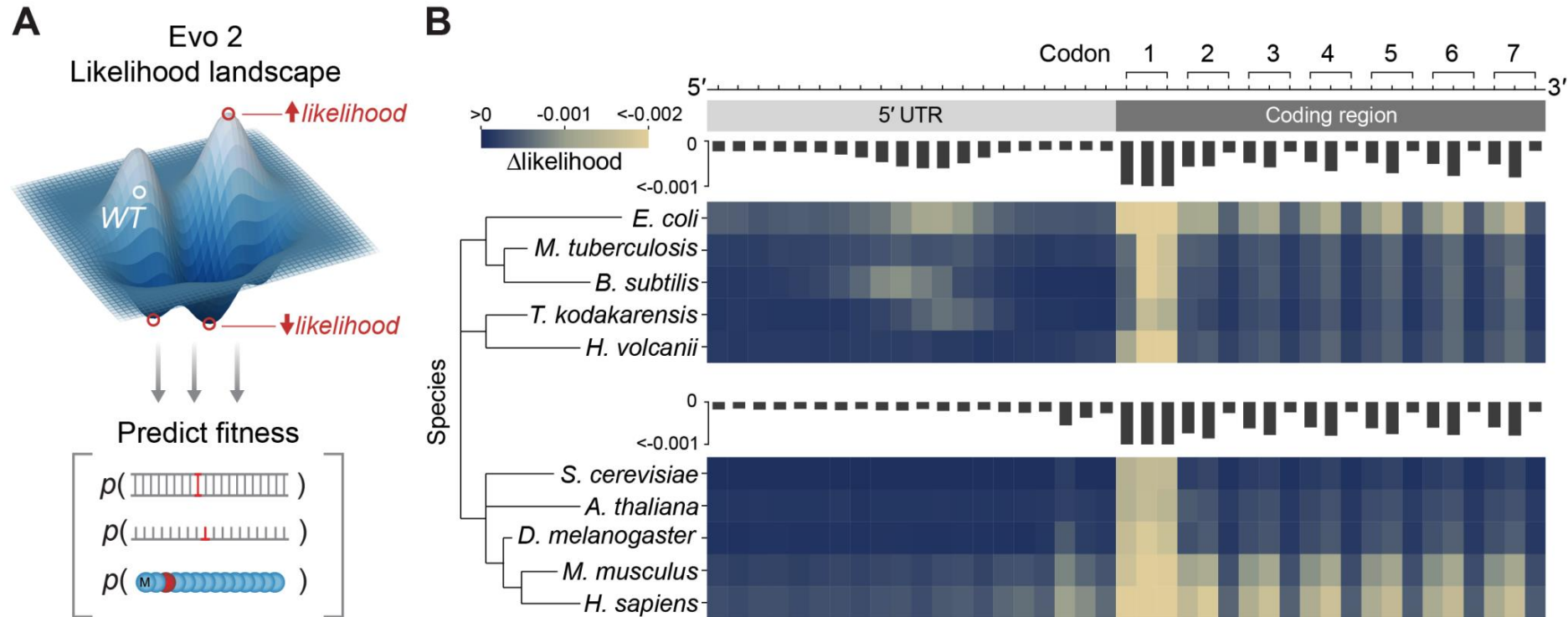


Training data

- OpenGenome2 dataset
- 113k prokaryotic genomes (357B bp)
- 15,032 eukaryotic genomes (6.98T bp)
- Non-redundant metagenomes (854B bp)
- Euk. organelle genomes (2.82B bp)
- Euk. coding sequences (602B bp)
- 8.8T bases total
- Data has no labels; **self-supervised** model

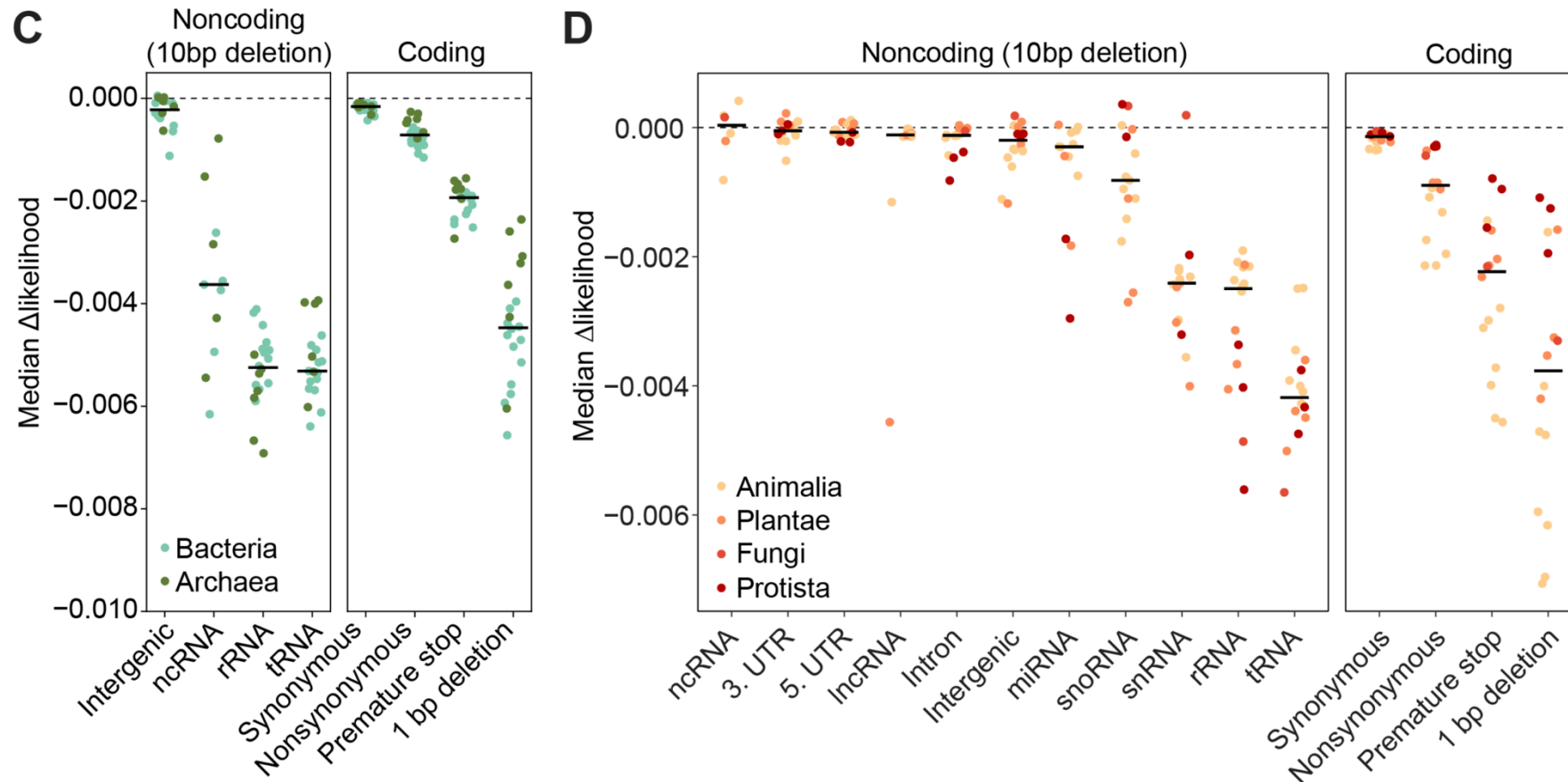


Evo2 learns coding syntax without any labels



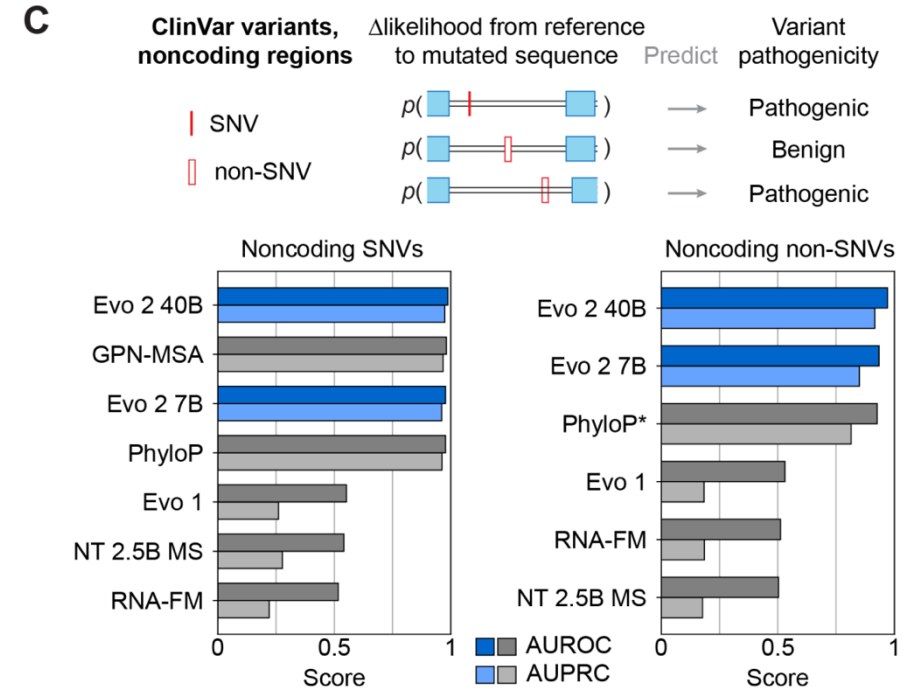
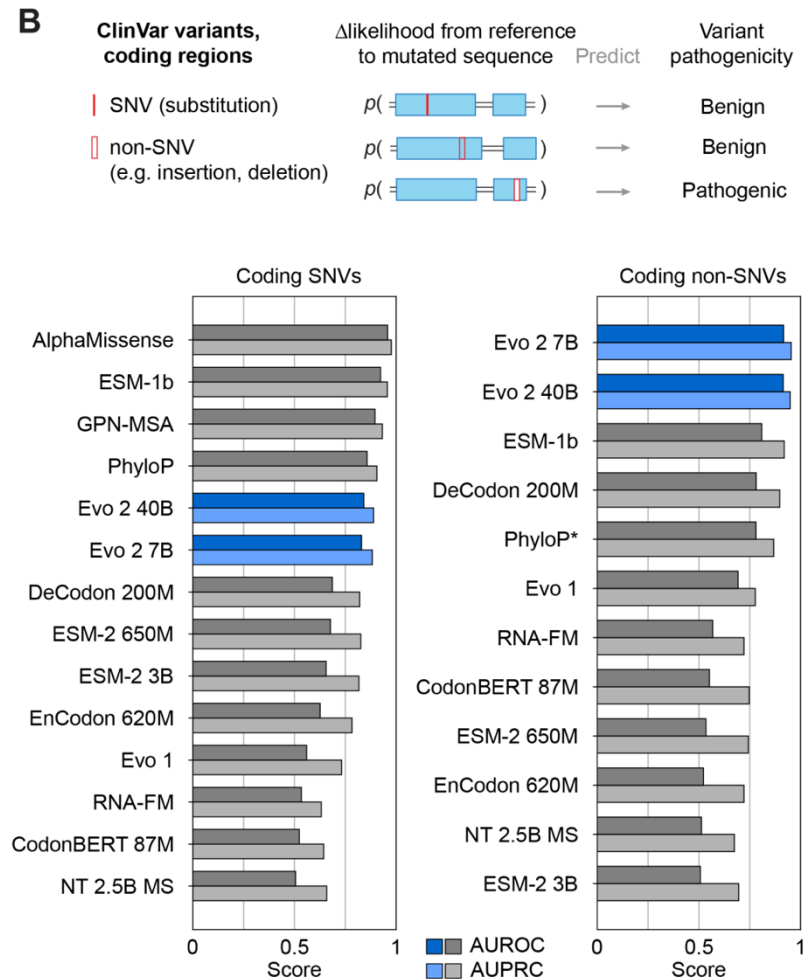
- **Mutate bases** at different positions and measure **change** in sequence **likelihood**
- Learns that mutations in the **1st codon base** are **very impactful** / unlikely
- Learns that mutations in the **3rd codon base** (wobble position) are **less impactful**

Evo2 learns mutation significance w/ no labels



- **Implicit** understanding that some mutations are more impactful than others

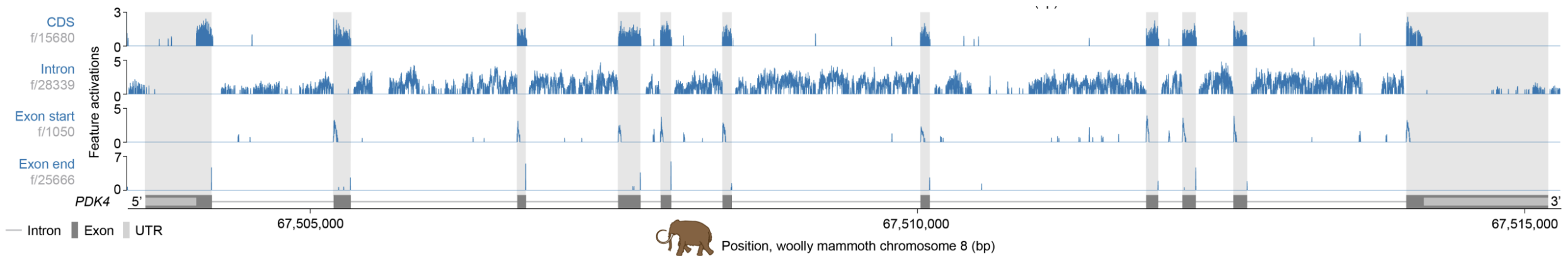
Good Evo2 zero-shot performance on ClinVar



- Uses **sequence likelihood** as a proxy for **variant effect**
 - Loss-of-function variants tend to be pathogenic
- 15k coding, 38k noncoding variants
- Competes with protein LMs and **exceeds** DNA LMs

Interpretable features can annotate genomes

- Use four features to annotate the woolly mammoth genome
- Introns, exons, and coding regions



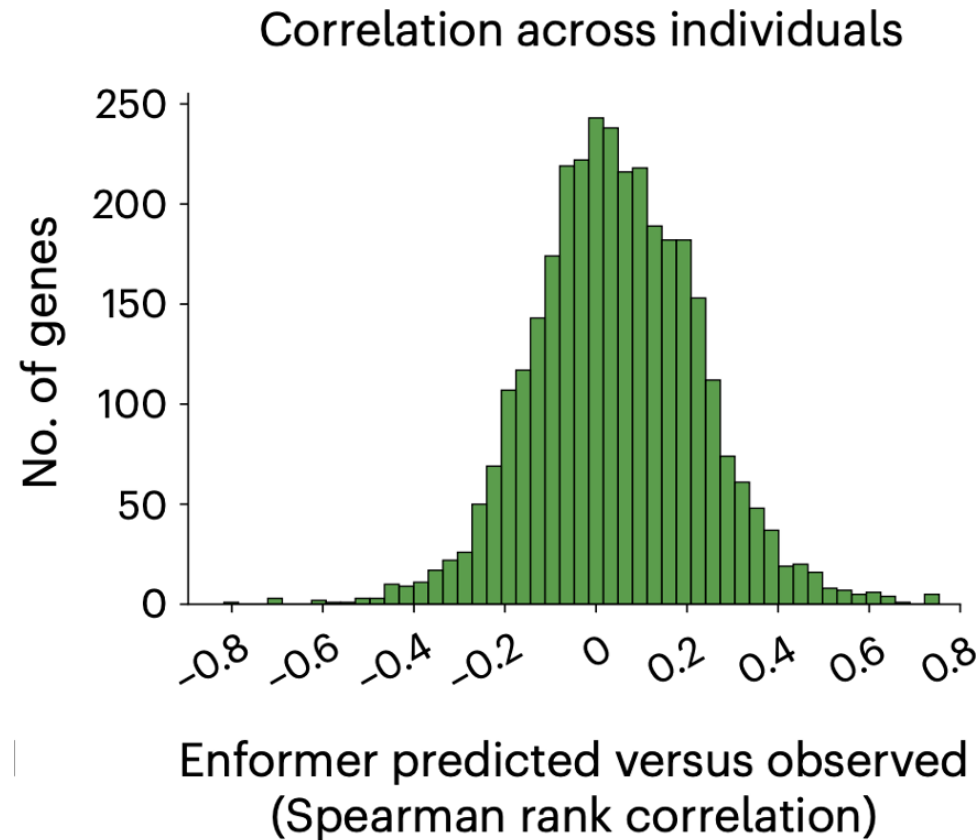
Enformer vs. Evo 2

- Represent **two paradigms** in sequence-to-function modelling
- Both have inspired a wide range of follow-up studies

	Enformer	Evo 2
Training strategy	Supervised	Self-supervised
Training genomes	Human + mouse	Thousands of organisms
Bases of training sequence	6 billion	9.3 trillion
Input context window	197 Kbp	1 Mbp
Parameters	250 M	40 B
Text processing module	Transformer	Striped Hyena 2

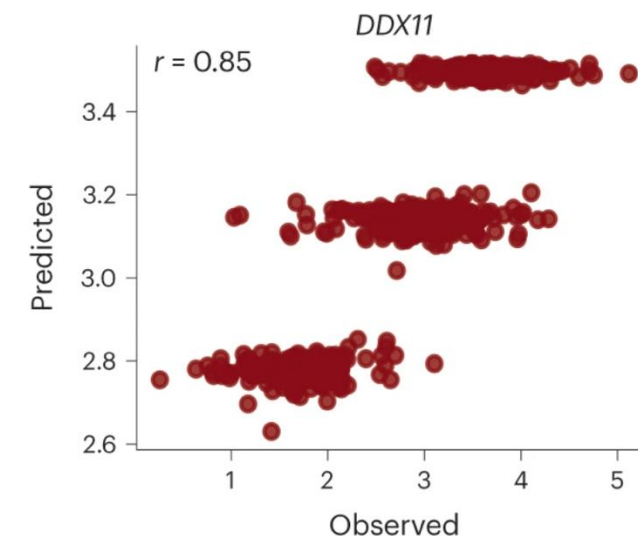
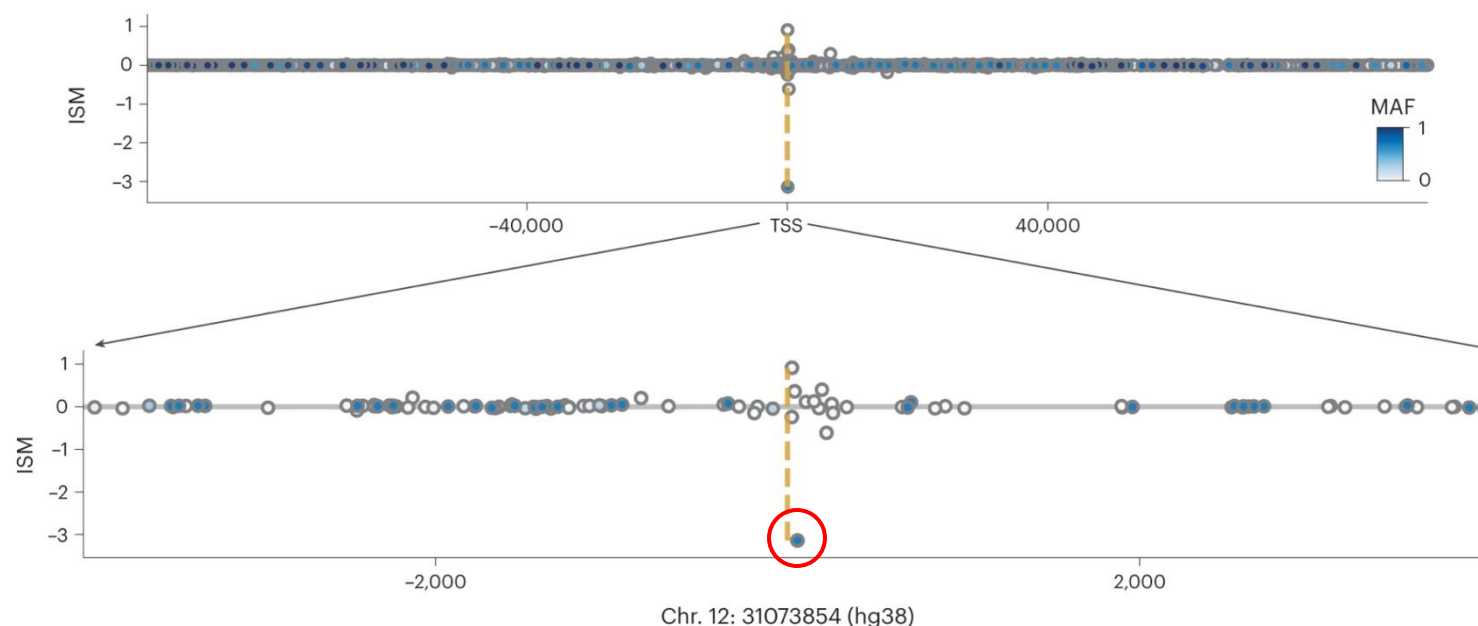
Is there room for improvement?

- Yes!



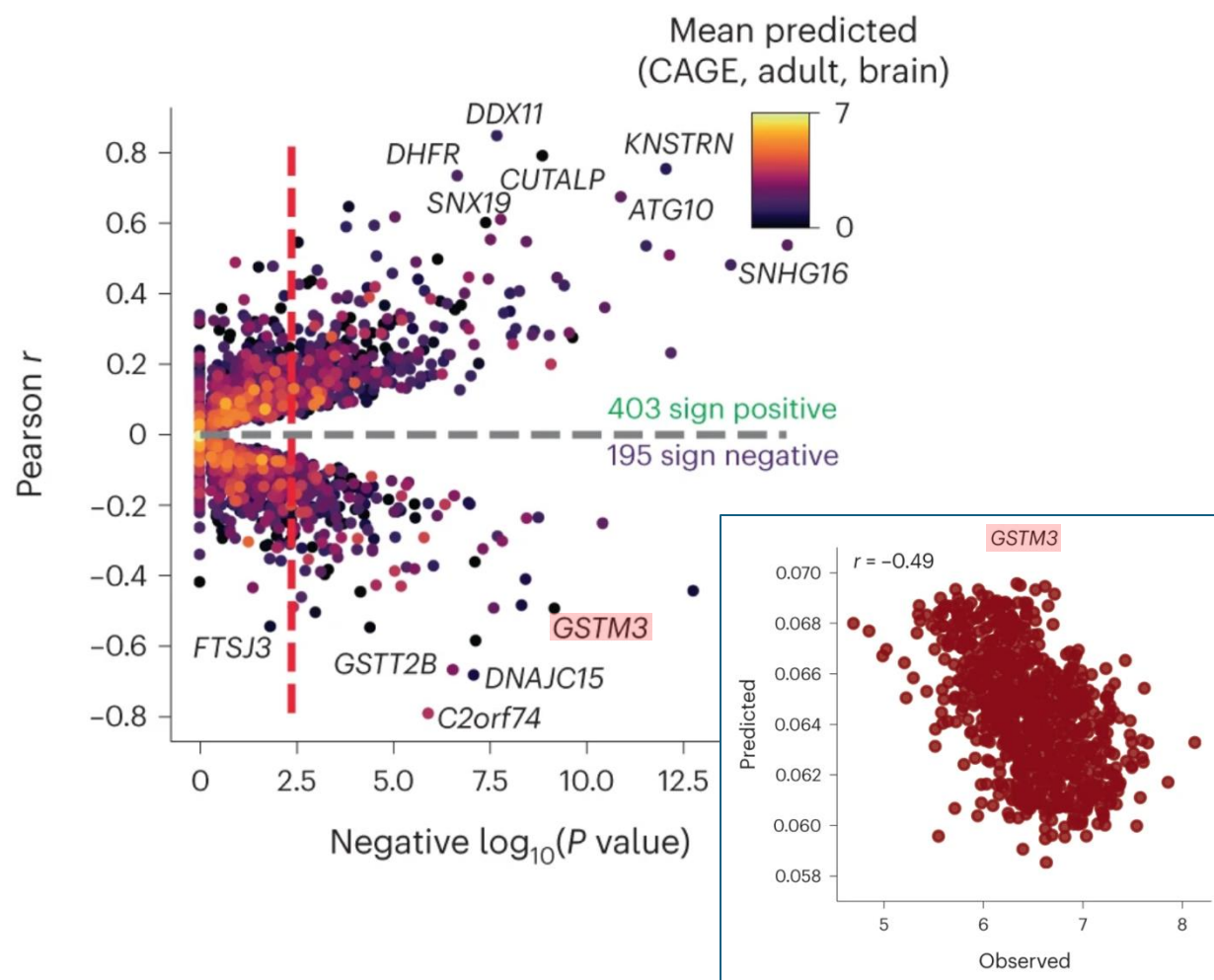
Using LLMs for variant eQTLs

- Predict **expression** directly from **sequence**
- In theory, can resolve linkage disequilibrium



- Enformer can use a **single CNV** out of many candidates to predict eQTL
- Selects same CNV identified by fine-mapping with Susie

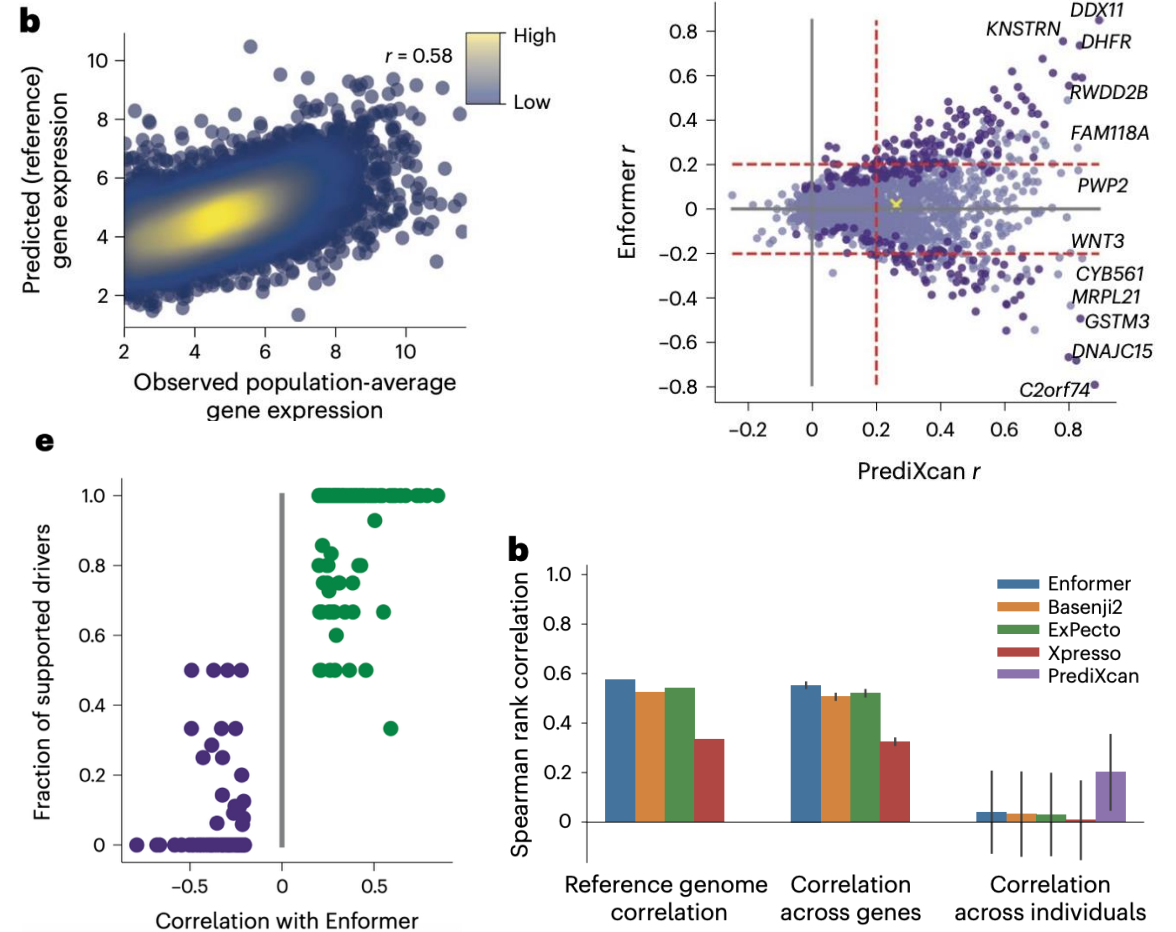
LLMs do not truly understand variants



- Previous slide is the **exception**, not the rule
- Overall, LLMs know **if** gene expression changes from a mutation but not **how** it changes
- Flip a coin as to whether expression goes up or down

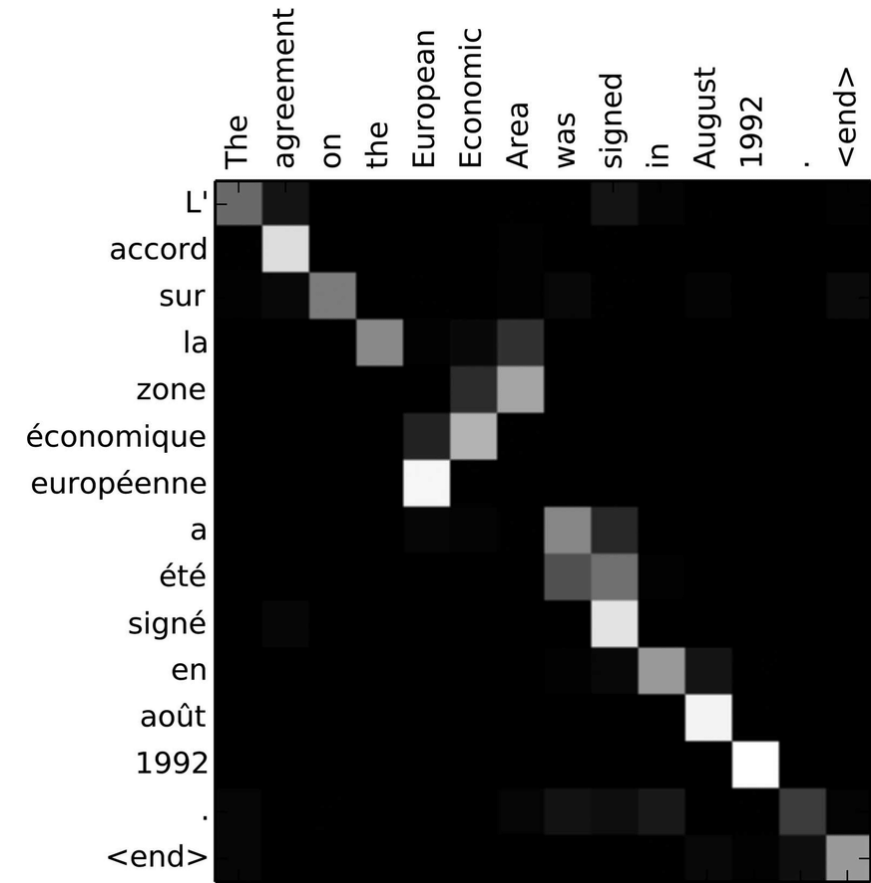
LLMs do not truly understand variants pt. 2

- Enformer is trained on the **reference** genome, not any **individual** sequences
- Top left: Enformer predicts population-average gene expression
- Bottom left: Enformer being right depends on finding causal SNVs
- Top, bottom right: linear model PrediXcan (purple) outperforms Enformer on individual-level predictions



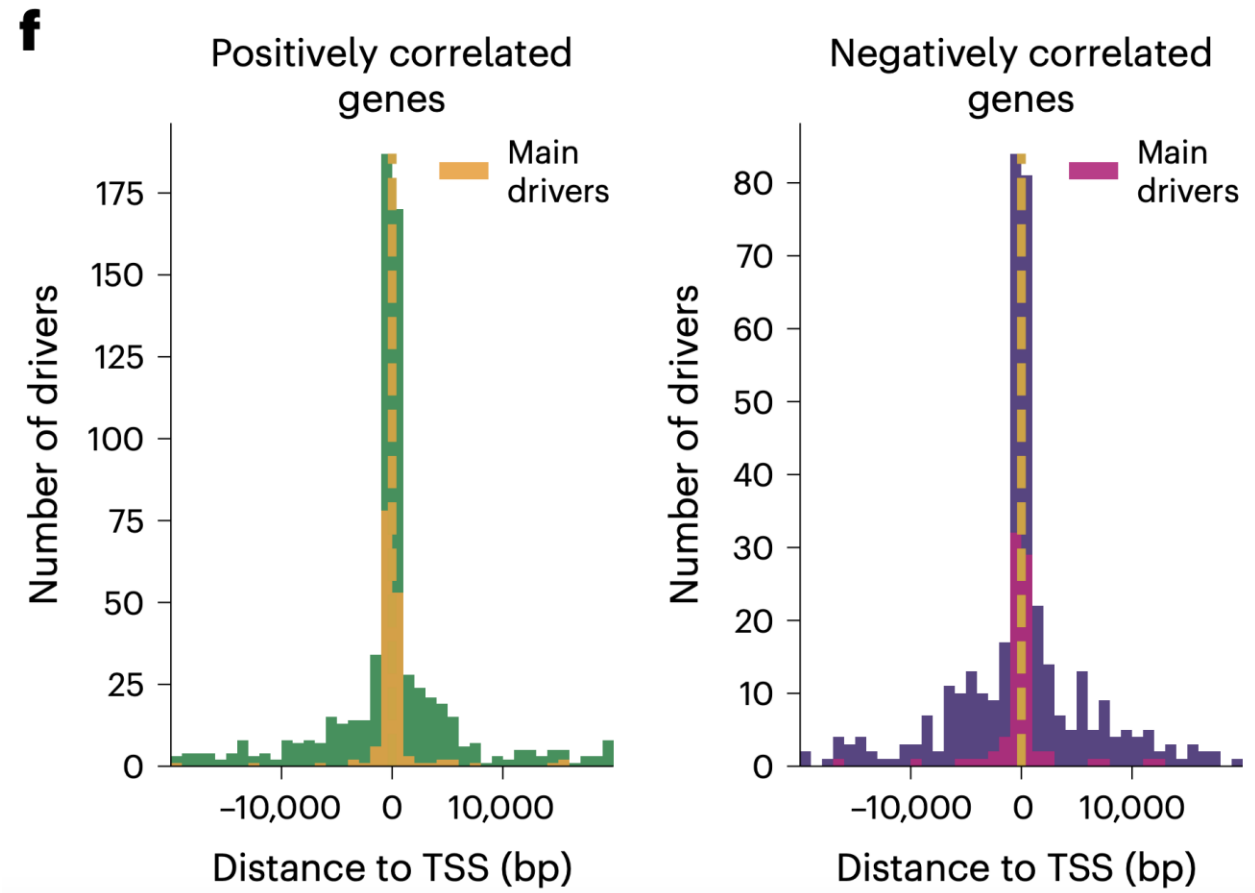
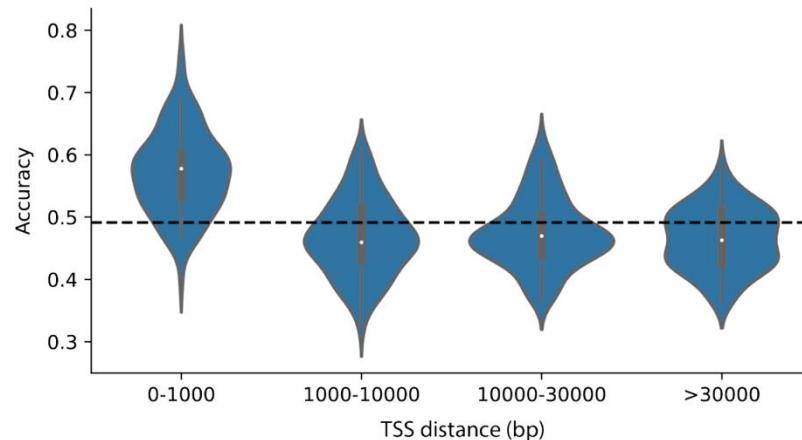
Expanding to longer inputs

- Genomic sequences are **long**, relevant info may be **far apart**
- Attention is a **quadratic** operation
 - 1K tokens = 1M operations
 - 1M tokens = 1T operations
- Current methods to address this:
 1. Dilation / downsampling
 - e.g. Enformer dilates via convolutions and pooling, losing resolution
 2. Mamba-style text module
 - e.g. Evo-2

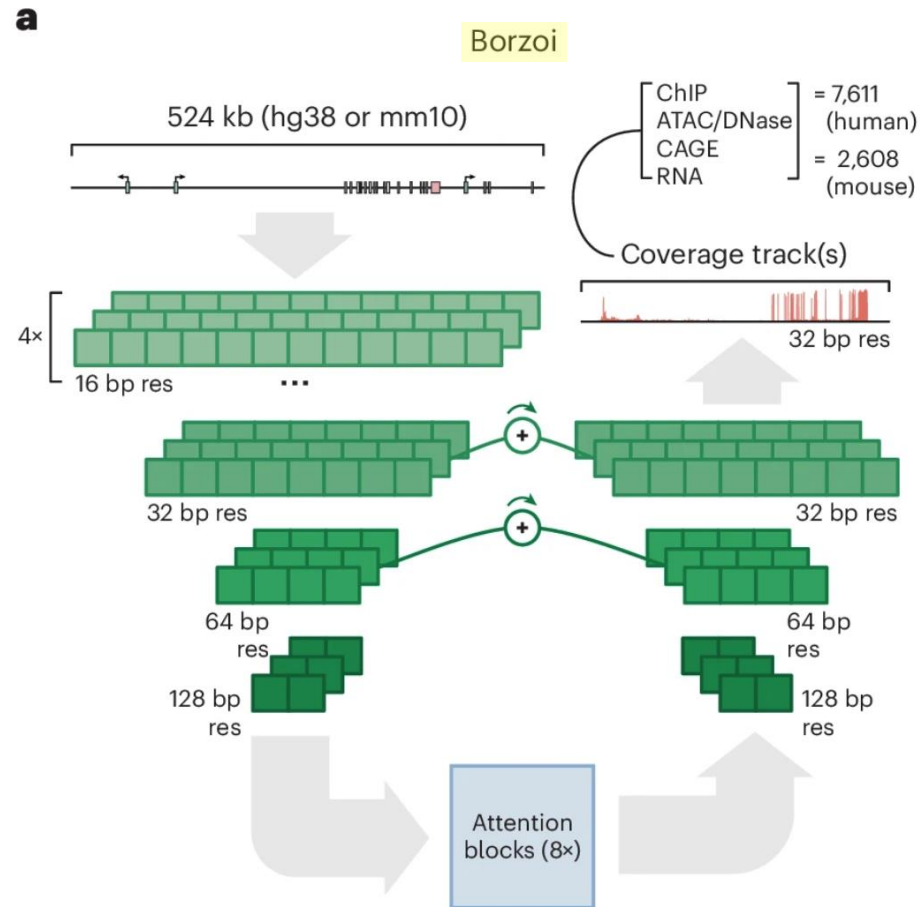


Making full use of long inputs

- Enformer generally finds drivers near the **transcription start site**
- Model is most **accurate** on variants near TSS



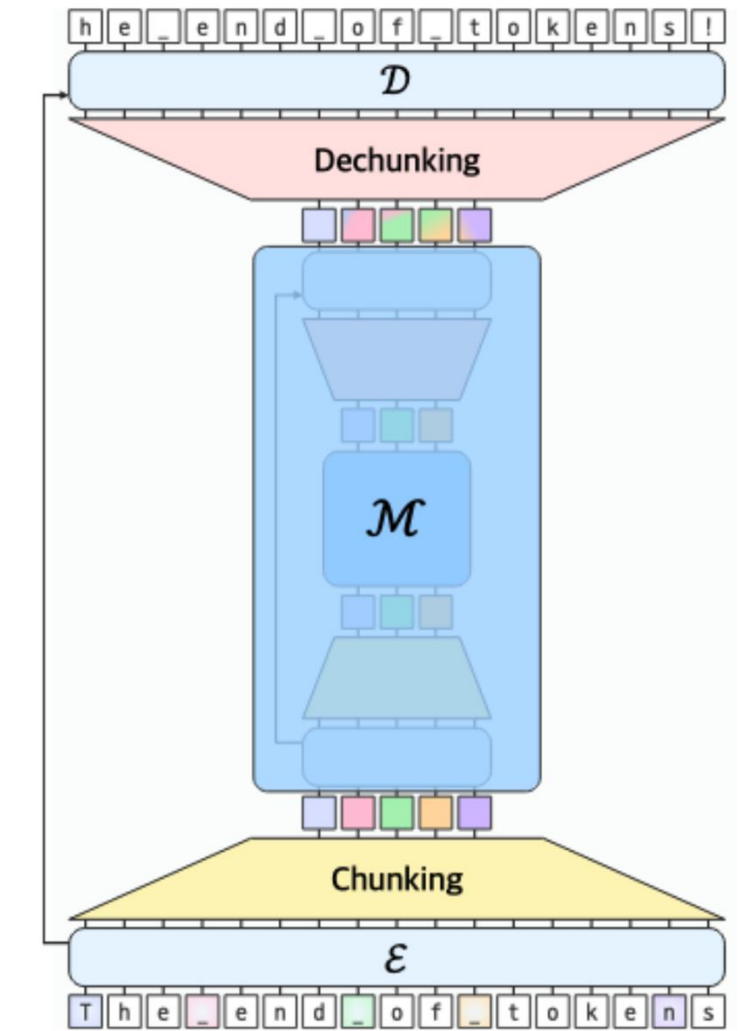
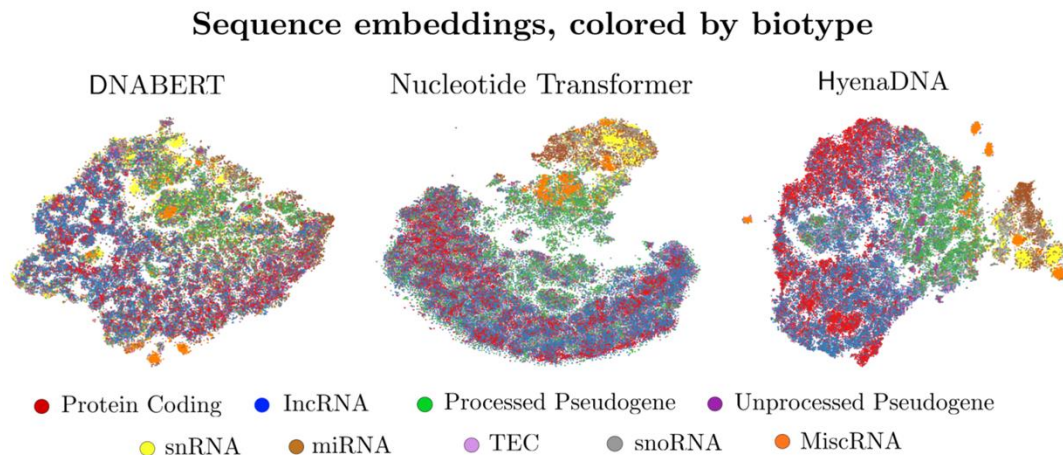
Better model architectures



- Borzoï's **updated architecture** over Enformer led to **performance gains**
- Frankly, their model is still **simple**
- Other ideas include:
 - Text processing evolution (attention -> mamba -> hyena -> stripedhyena)
 - Pooling modules (e.g. bag-of-mer)
 - Positional encodings (RoPE, ALiBi)
 - Algorithmic advances (e.g. alternating attention)

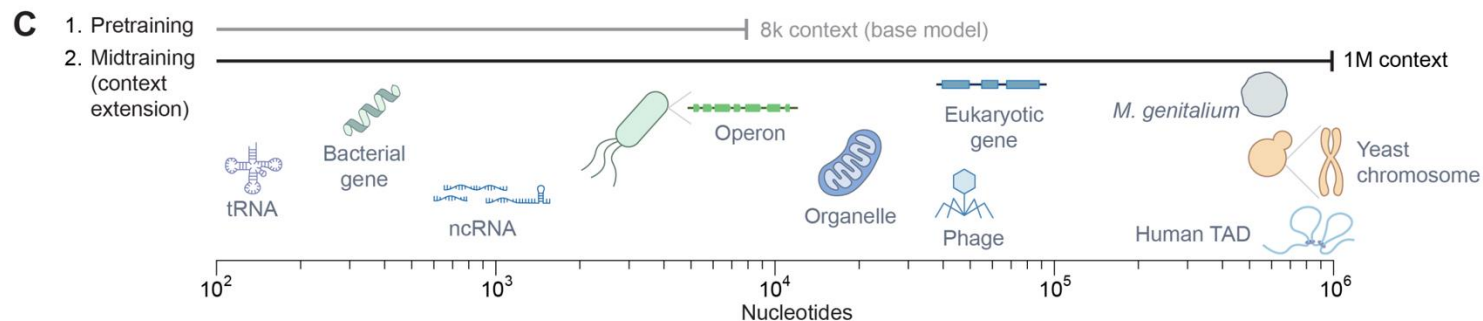
Better tokenizations

- How to **input** DNA sequences into the model?
- Kmer vs. single-nucleotide vs. byte-pair **tokens**
- **Unsupervised** tokenization (H-net)
- **Image-based** inputs (DeepSeek-OCR)

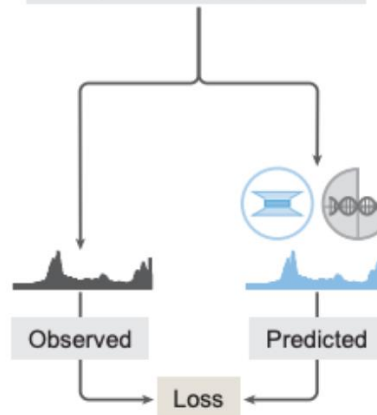
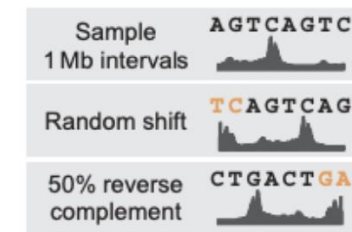


Better training regimes

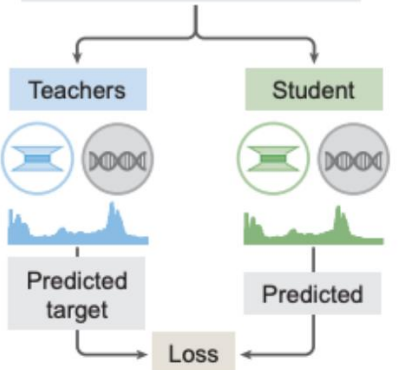
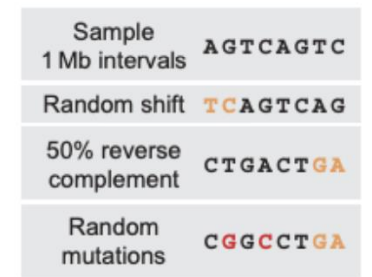
- More creative ways to **train** the model
- Evo-2 mostly trains on short sequences, then **extends** to long ones
- AlphaGenome trains a big **teacher** model, then trains small **student** models to **mimic** the teacher at **lower cost**



b Pre-training process

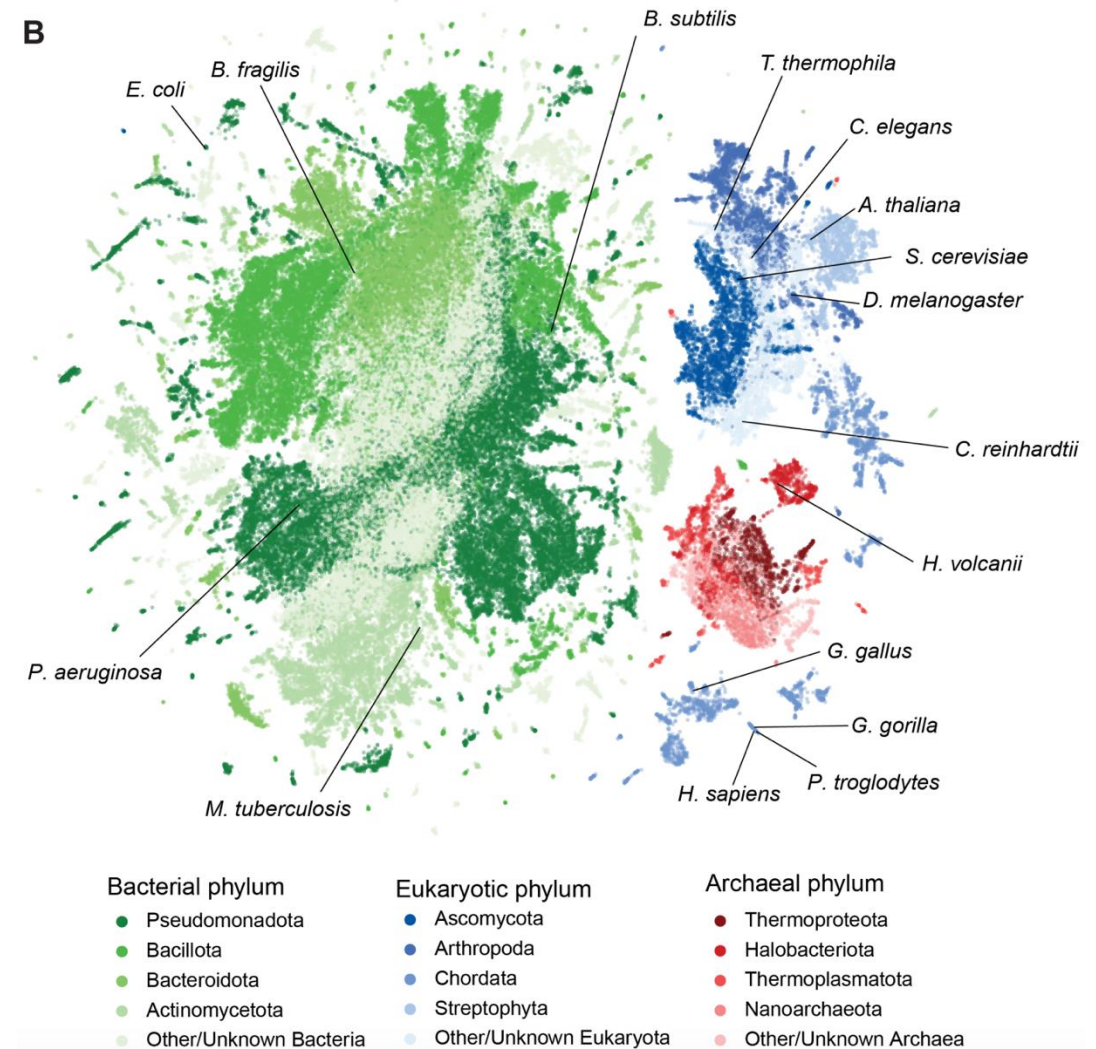


C Distillation process



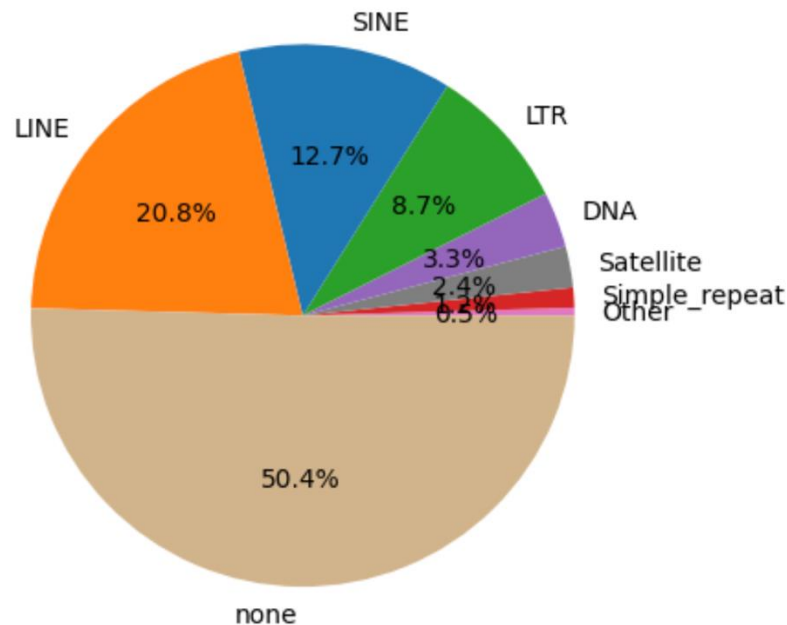
Better training datasets

- Evo-2's training set (right) vs. the rest of the field (pick 1-2 blue dots)
- ChatGPT trains on **trillions** of samples, why shouldn't we?
- Self-supervised models can train on **more data**
 - Not limited to labeled data



More efficient training datasets (my research!)

Composition of human genome by each repeat type



- Eukaryotic genomes are **repetitive**
- Repetitive data **worsens** LLMs

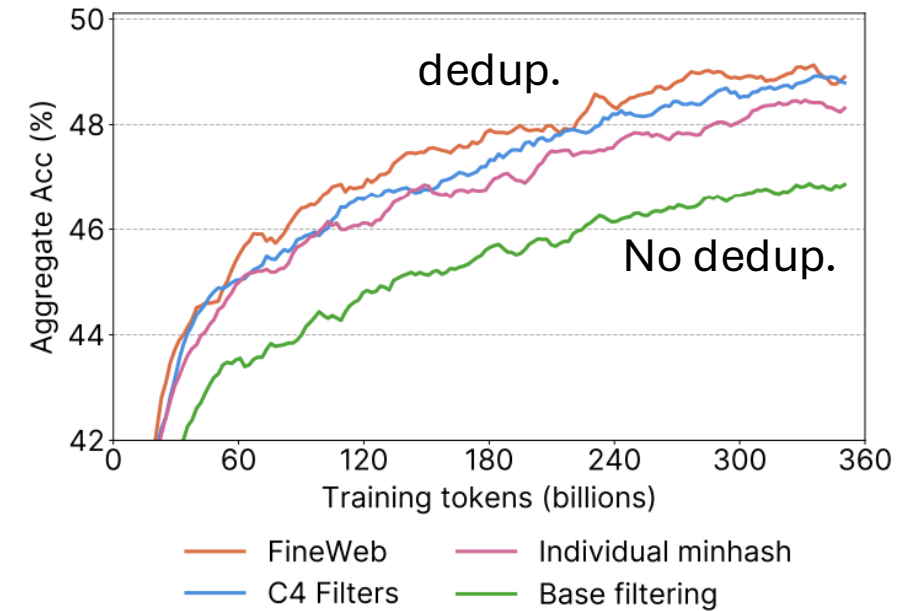
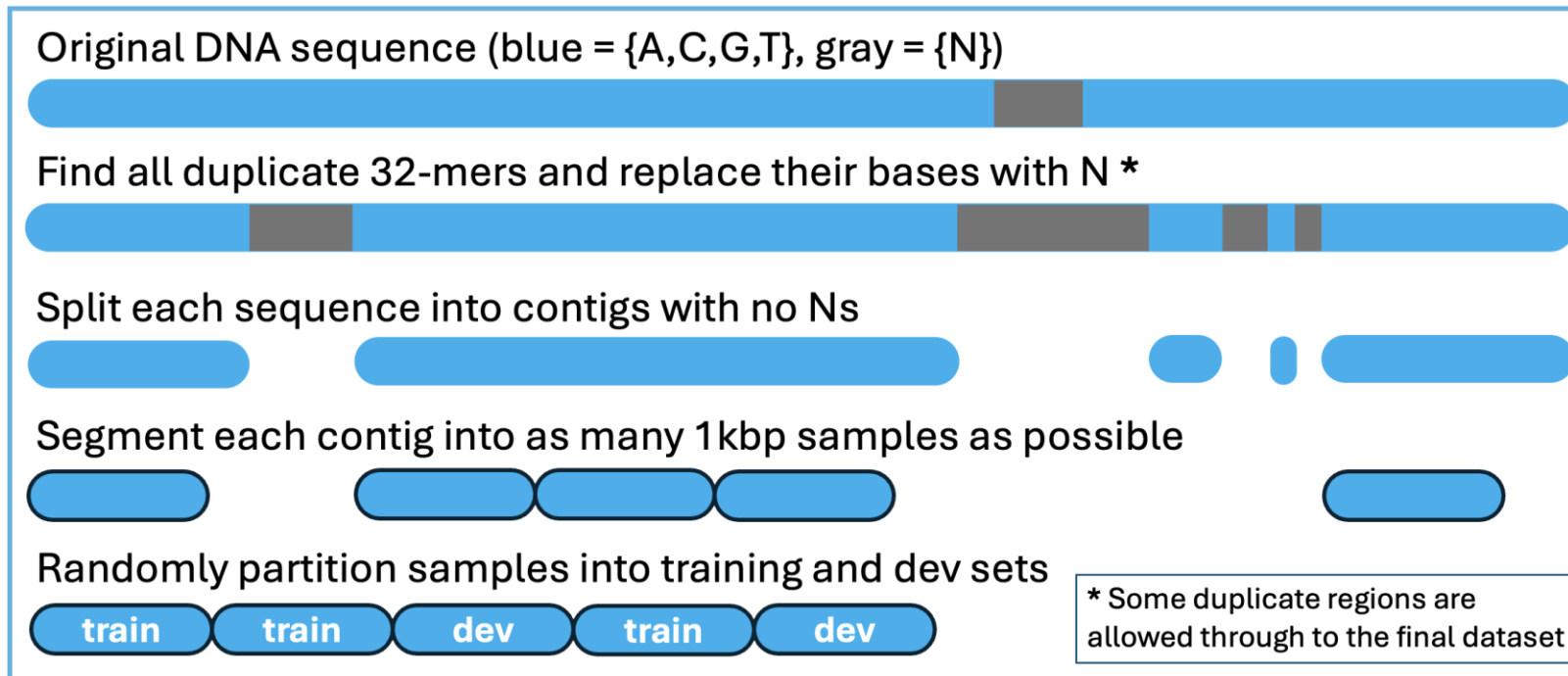


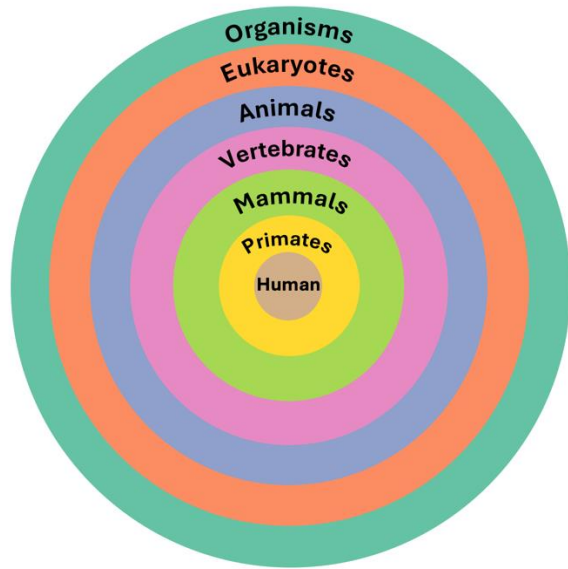
Figure 9: **Each processing step in FineWeb provides a performance uplift.** Compared to the base filtering (Section 3.3), applying individual-crawl Min-Hash deduplication (Section 3.4) the C4 filters (Section 3.5), and our additional heuristic filters (Section 3.6) each improve performance.

My research: genome deduplication



- Developed a method to **deduplicate** genomic data
- Removes any repeated **32-base** sequences from a training dataset

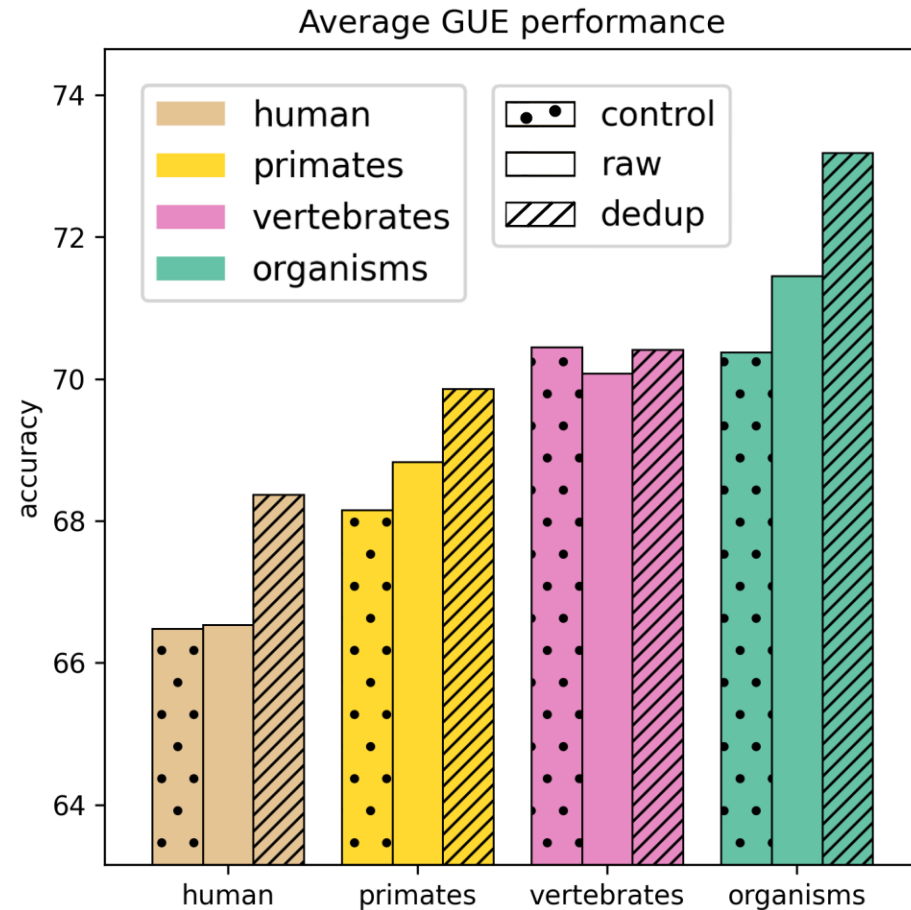
My research: genome deduplication



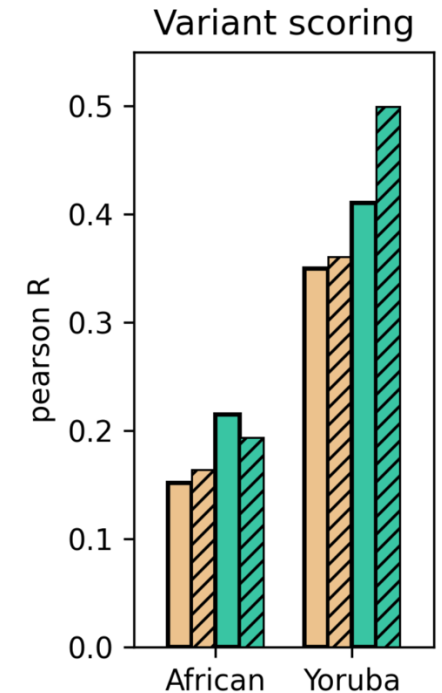
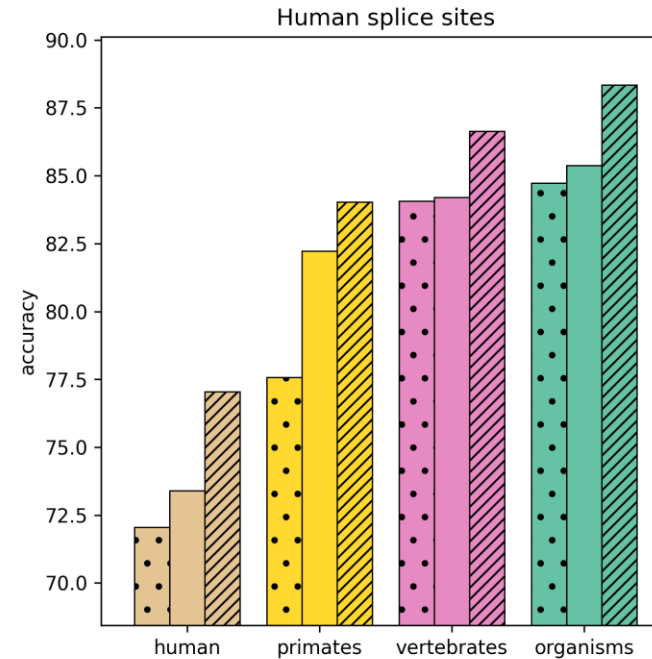
Dataset	Genomes	Raw bases (B)	Dedup. bases (B)	% dedup.	Training steps (K)
Human	1	3.0	1.5	50.0 %	50
Primates	3	8.4	2.5	70.2 %	100
Mammals	7	18.0	6.6	63.5 %	300
Vertebrates	12	25.4	9.2	63.8 %	400
Animals	21	29.9	10.2	65.7 %	500
Eukaryotes	34	30.3	10.6	65.1 %	500
Organisms	134	30.6	10.9	64.5 %	500

- Created **deduplicated datasets** for a variety of genome collections
- **Pretrained** a DNABERT-2-type model on each dataset
- **Fint-tuned** each model on **biologically-relevant** tasks

My research: genome deduplication



Training on **deduplicated** data leads to better **performance** on biologically relevant tasks!



Summary thoughts

- We have already seen **AI** and **deep learning** transform our world
- Speaks for itself that Large Language Models are powerful
- **DNA**, like English or Mandarin or Python, is a **language**
- LLMs may be able to transform our **understanding** of DNA



Summary thoughts

- Genomic LLMs have shown great promise in understanding the **grammar** of DNA
- Already applicable to predicting **transcription factor** binding, **chromatin** accessibility, **splice sites**, and more
- Constantly improving at **variant** effect prediction
- Much **more work** to be done before the models are fully usable
- **Supervised** vs. **self-supervised** is the defining debate in the field

What does a DNA variant do?

Method	Drawback(s)
Query it in a database	The variant might not exist in a database The variant might act differently under different conditions
Assume loss-of-function of nearby gene	Not always an accurate assumption Function of nearby gene may not itself be understood
Genome-Wide Association Study (GWAS)	Not applicable for variants that only exist in a single person Underpowered for rare variants that only occur in a few people Relies on undersized datasets containing a small number of SNPs Requires rigorous statistical testing Can only model one variant at a time Assumes additive linear effects of variants Difficult to adapt to a polygenic/omnigenic model of traits In practice, only explains a fraction of heritability
Deep learning	Computationally expensive Not yet sophisticated enough to give satisfying answers to all questions

Deep Learning in Genomics Journal Club

- Discussion group for students interested in genomic deep learning
- We meet every other Tuesday at 12pm
- Participation is optional, feel free to just sit and listen
- Next meetings 11/4 and 11/18
- Email me if interested! All are welcome

